A certified continuation machine for Mahler measure identities at CM points, with twelve new proofs of conjectures of Samart

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Declaration on the use of AI tools

The research reported in this article — including the computational exploration, the discovery of the proof strategy, the machine-certified verifications, and the preparation of the manuscript — was carried out by the author with the assistance of the AI system *Kimi* (Moonshot AI). All mathematical content, including every proof and every certified computation, has been checked and verified by the author, who takes full responsibility for the correctness and integrity of the article. All certification scripts are available for independent verification (see Section 2.5).

Abstract

We axiomatize the differential-comparison continuation method of [5] into a general machine for proving Mahler measure identities at CM points: a modular parametrization with rigorous tail bounds, a single-valued holomorphic Mahler differential on the complement of the critical image, a propagation lemma that replaces all monodromy and universal-cover arguments, interval-arithmetic path certification, and an exact three-track CM evaluation. Applied to Samart's family $n_4(s) = 4\mathrm{m}\big(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz\big)$, the machine proves twelve further entries of his 2015 table [2, Table 6]: the conjugate pairs at $s = 8292456 \pm 3132675\sqrt{7}$ (whose two identities are proved separately, not merely as a sum), $s = 3656 \pm 2600\sqrt{2}$, $s = -144$, the conjugate pair $s = 143208 \pm 101574\sqrt{2}$ attached to $\mathbb{Q}(\mathrm{i})$, the two values $s = 1207368 + 853632\sqrt{2} \pm 697680\sqrt{3} \pm 493272\sqrt{6}$ with matching signs attached to the class-number-two field $\mathbb{Q}(\sqrt{-6})$ — the first Mahler measure identities in the n_4 family proved at *non-degenerate* CM points of class number two (Samart's $s = 2304$ [1], attached to $\tau = \sqrt{-6}/2$, is the degenerate value $s_4(\mathrm{i}\sqrt{6}/2) = s_4(\mathrm{i}\sqrt{6}/6)$), where the lattice sums involve two newforms and a genus character; the value $s = (-192303 - 85995\sqrt{5})/2$ attached to the class-number-two field $\mathbb{Q}(\sqrt{-15})$; and the pair $s = -893952 \pm 516096\sqrt{3}$ attached to $\mathbb{Q}(\sqrt{-21})$ — the first Mahler measure identities proved at CM points of class number four (the $\mathbb{Q}(\sqrt{3})$ pair attached to $\mathbb{Q}(\sqrt{-21})$), where the lattice sums involve four newforms and the genus group $(\mathbb{Z}/2)^2$. New methodological ingredients include: the Fricke ratio as the only non-vacuous level/root-number discriminator (the "functional-equation residue" test is vacuous for root number $+1$ at every level); a genuine twist-level trap ($g_{16} \otimes \chi_8$ has level 64, not 32, and the wrong level silently scales the L-value); a genus-character decomposition of the lattice sums in class number two; and the discovery that Samart's applicability boundary $\{\mathrm{Im}\tau = 1/\sqrt{2}\}$ is not the topological boundary of the continuation region V_4. Every identity is certified by an exact algebraic track (rational arithmetic only, up to the quoted CM-theory inputs disclosed in Section 2.5), rigorous final-identity interval locks (half-widths down to 9.1×10^{-53}), and independent 50–60-digit numerical cross-checks; all certification scripts are public. Beyond Table 6, the evaluation track of the machine

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produces six new identities at Heegner points (discriminants $D = 11, 19, 27, 43, 67, 163$), stated as conjectures with 60-digit evidence and exact lattice-derived coefficients; they require a new ray-class constant type outside Samart's library. We close with an umbrella conjecture (a CM specialization conjecture with discriminant structure, and a Tamagawa-controlled denominator conjecture) organizing the twelve proved and six conjectured identities. Finally we score the five remaining open entries of Table 6 against the machine's five stations: all five are blocked by genuine obstructions — a two-dimensional critical image or a wrong-sheet Eisenstein–Kronecker series — which we analyze precisely.

Keywords: Mahler measure · modular forms · complex multiplication · special values of L-functions · rigorous interval arithmetic

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1 Introduction

1.1 The question

Let $\mathrm{m}(P)$ denote the logarithmic Mahler measure of a Laurent polynomial and let

$$n_4(s) := 4\,\mathrm{m}\big(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz\big), \qquad s \in \mathbb{C},$$

the value being independent of the choice of the fourth root (Section 2). In [1, 2] Samart expressed $n_4(s)$, for s in the image of a modular parametrization $s_4(\tau)$, as an Eisenstein–Kronecker (EK) series, proved the identity for pure imaginary τ with $\mathrm{Im}\tau \geq 1/\sqrt{2}$, and conjectured on numerical evidence the twenty-nine L-value evaluations of his Table 6 [2]: each attaches to a CM value of s a rational linear combination of L-values of CM weight-three newforms and of quadratic Dirichlet characters.

The table is now largely, but not entirely, resolved. Eleven entries were proved previously: the exterior values $s = 256$ (Rogers [8]), $s = 648, 2304, 20736, 614656$ (Samart [1]), $s = 26856 \pm 15300\sqrt{3}$ (Samart's own Theorem 3.1 [2]), the outside-the-critical-locus value $s = -3969$ (Fei [4]), and $s = -1024, -12288, -82944$ (Guo–Peng–Qin [11]). One entry is *refuted*: the conjecture $n_4(81) = 40M_7$ fails numerically as a statement about the true Mahler measure, the conjectured value being the evaluation of the EK series at the attached CM point on a wrong sheet [6]. He–Ye [3] resolved the companion family n_3 completely; the n_2 family was completed in [5, 10, 11]. Fei [4] proved, for all rational singular moduli in twenty-three families, identities at the level of the real part of the *holomorphic* Mahler measure $\mathrm{Re}\,\widetilde{\mathrm{m}}$; at parameters inside the critical locus the passage from $\mathrm{Re}\,\widetilde{\mathrm{m}}$ to the genuine Mahler measure is not automatic — a point stressed already by Rodriguez Villegas [9] — and it is precisely this passage that the continuation machine of [5] and of the present paper performs. Altogether, twenty-three of the twenty-nine entries are now proved — eleven previously, twelve in the present paper (Theorems A–K) — one is refuted, and five remain open.

Remark 1.1 (The status of $s = -3969$). For the entry $s = -3969$ above, [4] proves the identity for $\mathrm{Re}\,\widetilde{\mathrm{m}}$ only, and the determination of the region off which $\mathrm{m} = \mathrm{Re}\,\widetilde{\mathrm{m}}$ is left open there (his Remark 2.4); by Section 10 below, avoiding the critical image alone does not suffice (the wrong-sheet phenomenon). The missing step is supplied by the present machine: the attached point $\tau = (1+\sqrt{-7})/2$ lies on the vertical segment $\mathrm{Re}\tau = 1/2$, $0.702 \leq \mathrm{Im}\tau \leq \sqrt{21}/2$, which `n5_line_cert.py` certifies inside the good component W by interval arithmetic (public repository; the same certificate covers the CM points of Theorems 8.2–8.5, see Section 8). Hence Theorem 2.10 applies and the $s = -3969$ evaluation holds for the genuine Mahler measure.

The purpose of this paper is threefold:

(i) to axiomatize the differential-comparison continuation method of [5] into a general, family-independent machine (Section 2), with an explicit certifier architecture whose trusted base consists of pure mathematics, documented interval-arithmetic semantics, and public source code (Section 2.5);

(ii) to apply the machine to twelve further entries of Samart's Table 6 (Theorems A–F below and Theorems G–K of Section 8), covering every regime that occurs: deep pure-imaginary interior points (A, C, D, E), a non-pure-imaginary interior point (B), the first class-number-two CM points (E, H, I), a point on Samart's applicability boundary that turns out to be a strict interior point of the continuation region (F), and the first class-number-four CM points (J, K);

(iii) to take stock: to score the five remaining open entries of Table 6 against the machine's checklist (§1.3) and to describe precisely where and why the method stops (Section 10).

1.2 Results of this paper

Throughout, $d_k := L'(\chi_{-k}, -1)$ for the odd quadratic Dirichlet character χ_{-k} of conductor k, and M_k or $M_k^{(i)}$ denotes $L'(g_k, 0)$, resp. $L'(g_k^{(i)}, 0)$, for the indicated weight-3 CM newform(s); a superscript tw denotes the L'-value of an explicitly identified quadratic twist.

Theorem 1.2 (Theorem A: the $\sqrt{7}$ pair)**.** *Let $M_7 := L'(g_7, 0)$ with $g_7(\tau) = \eta(\tau)^3\eta(7\tau)^3$ (LMFDB* 7.3.b.a*) and $M_7^{\mathrm{tw}} := L'(g_7 \otimes \chi_{-4}, 0)$. Then*

$$n_4\big(8292456 + 3132675\sqrt{7}\big) = \frac{5}{28}\big(4M_7^{\mathrm{tw}} + 224M_7 + 32d_4 + 7d_7\big), \tag{1}$$

$$n_4\big(8292456 - 3132675\sqrt{7}\big) = \frac{5}{14}\big(4M_7^{\mathrm{tw}} - 224M_7 + 32d_4 - 7d_7\big). \tag{2}$$

Theorem 1.3 (Theorem B)**.** *With $M_{12} := L'(g_{12}, 0)$, $g_{12}(\tau) = \eta(2\tau)^3\eta(6\tau)^3$ (LMFDB* 12.3.c.a*),*

$$n_4(-144) = \frac{10}{3}\big(4M_{12} + d_3\big).$$

Theorem 1.4 (Theorem C)**.** *With $M_8 := L'(g_8, 0)$, $g_8(\tau) = \eta(\tau)^2\eta(2\tau)\eta(4\tau)\eta(8\tau)^2 \in S_3(\Gamma_0(8), \chi_{-8})$, and $M_8^{\mathrm{tw}} := L'(g_8 \otimes \chi_8, 0)$, where $\chi_8 = (2/\cdot)$ is the even primitive character of conductor 8,*

$$n_4\big(3656 + 2600\sqrt{2}\big) = \frac{5}{8}\big(4M_8^{\mathrm{tw}} + 28M_8 + 4d_4 + d_8\big).$$

Theorem 1.5 (Theorem D)**.** *With $M_{16} := L'(g_{16}, 0)$, $g_{16}(\tau) = \eta(4\tau)^6 \in S_3(\Gamma_0(16), \chi_{-4})$, and $M_{16}^{\mathrm{tw}} := L'(g_{16} \otimes \chi_8, 0)$ (level 64 — see §5.4),*

$$n_4\big(143208 + 101574\sqrt{2}\big) = \frac{5}{16}\big(4M_{16}^{\mathrm{tw}} + 20M_{16} + 9d_4 + 4d_8\big).$$

Theorem 1.6 (Theorem E: class number two)**.** *Let $K = \mathbb{Q}(\sqrt{-6})$ (class number 2), and let $g_{24}^{(1)}, g_{24}^{(2)}$ be the two newforms of $S_3(\Gamma_0(24), \chi_{-24})$, theta series of the two grössencharacters of K of type $(2, 0)$ (LMFDB* 24.3.h.a *and* 24.3.h.b*). With $M_{24}^{(i)} := L'(g_{24}^{(i)}, 0)$ and $M_{24}^{(i),\mathrm{tw}} := L'(g_{24}^{(i)} \otimes \chi_{-8}, 0)$ (level 96),*

$$n_4\big(1207368 + 853632\sqrt{2} + 697680\sqrt{3} + 493272\sqrt{6}\big)$$
$$= \frac{5}{48}\Big(4M_{24}^{(1),\mathrm{tw}} + 4M_{24}^{(2),\mathrm{tw}} + 28M_{24}^{(1)} + 12M_{24}^{(2)} + 28d_3 + 24d_4 + 8d_8 + d_{24}\Big).$$

Theorem 1.7 (Theorem F: a boundary point that is interior)**.** *With the notation of Theorem 1.4,*

$$n_4\big(3656 - 2600\sqrt{2}\big) = \frac{5}{4}\big(4M_8^{\mathrm{tw}} - 28M_8 + 4d_4 - d_8\big).$$

The s-value is the conjugate of Theorem 1.4, but the attached CM point $\tau_4 = (1+\sqrt{-2})/2$ lies exactly *on Samart's applicability boundary* $\mathrm{Im}\tau = 1/\sqrt{2}$*, where his proof does not apply; the identity is proved by the continuation machine* (*Section 7*)*, which shows that τ_4 is a strict interior point of the continuation region V_4* (*certified margin* 0.11590).

Theorem 1.8 (The machine, informal)**.** (*Precise form: Theorem 2.10.*) *Let P_c be a tempered Laurent-polynomial family with a modular parametrization $c = c(\tau)$ and critical image $K \subset \mathbb{C}$, and let $E(\tau)$ be the Eisenstein–Kronecker expression for the holomorphic Mahler measure, valid on an open set $U \subset \mathbb{H}$. If* (i) *the period integral $\Omega(c)$ is single-valued and holomorphic on $\mathbb{C}\setminus K$,* (ii) *$c(\tau_0)$ is a CM parameter with $\tau_0 \in \overline{W}$, where W is the connected component of $\mathbb{H}\setminus c^{-1}(K)$ containing U, and* (iii) *the lattice sums underlying $E(\tau_0)$ evaluate exactly, then $\mathrm{m}(P_{c(\tau_0)})$ equals the corresponding L-value expression; the inclusion $\tau_0 \in \overline{W}$ is certifiable by rigorous interval arithmetic.*

All twelve identities are machine-certified. Concretely, each proof consists of an exact algebraic track (rational/algebraic-integer arithmetic only, no floating point), rigorous interval locks of the CM values and of the final identities (final-identity half-widths between 9.1×10^{-53} and 2.3×10^{-51}), and independent 50–60-digit numerical cross-checks; the scripts are listed in Section 2.5 and are publicly available (see the Data Availability statement). Two representative reference values:

$$\begin{aligned} M_8 &= 0.1353184954231269106150060654595395883532\ldots, \\ M_8^{\mathrm{tw}} &= 1.549084847095611486271658159444376281785\ldots \quad (\text{level } 32,\ w=+1), \end{aligned}$$

computed from the functional equations and cross-checked against direct lattice-sum evaluations; further values are collected in the application sections.

Two further outputs of the same machinery do not belong to Table 6 and are collected in Section 9: six new identities at the Heegner points $\tau_D = (1+\sqrt{-D})/2$, $D \in \{11, 19, 43, 67, 163\}$, and at $\tau_{27} = (1+3\sqrt{-3})/2$ (Conjecture 9.2), whose coefficients are produced exactly by the machine's evaluation track and which require a new ray-class constant type beyond Samart's M/d_k library (Remark 9.5); and an umbrella conjecture (Conjectures 9.6 and 9.7) organizing the twelve proved and six conjectured identities and predicting their discriminant and denominator structure.

1.3 The five remaining entries

Table 6 of [2] has twenty-nine entries: twenty-three are now proved (eleven previously known, plus the twelve of Theorems A–K), one is refuted ($s = 81$ [6]), and five remain open. The machine reduces a proof of each entry to five *stations* (Section 2 for definitions):

- **S1** *Parametrization*: a CM point τ_0 with $s_4(\tau_0) = s$ identified, at least numerically;
- **S2** *Critical image*: the position of $c(\tau_0)$ relative to the astroid disc Ast and the cross $[-4,4] \cup \mathrm{i}[-4,4]$;
- **S3** *Interior-or-boundary*: membership of τ_0 in $\overline{W}$, certified by a `cert2` path, or direct applicability of Samart's theorem;
- **S4** *`cert0` algebraicity*: an exact algebraic lock of the value $s_4(\tau_0)$;

S5 *P1 assemblability*: an exact CM evaluation of $\mathrm{EK}_4(\tau_0)$ in the relevant imaginary quadratic field.

Table 1 scores the five open entries against these stations.

Table 1: The five open entries of Samart's Table 6, scored against the five stations of the machine (the original numbering of the ten-entry list is kept; entries #1, #2, #6, #7, #8 are now proved as Theorems 8.1–8.5). "Outside/inside" refers to the astroid disc Ast; ζ abbreviates 1207368, $a = 853632$, $b = 697680$, $d = 493272$ in rows 3–4: $s = \zeta \pm a\sqrt{2} \pm b\sqrt{3} \pm d\sqrt{6}$ with the indicated signs.

#	s	τ_0	$\mathrm{Im}\tau_0$	S2: $c(\tau_0)$	S3: position	S4/S5 status	verdict
3	ζ, signs $(-+-)$	$\mathrm{i}\sqrt{6}/4$	0.612	outside Ast ($s > 256$)	*exterior*: EK series wrong-sheeted below $1/\sqrt{2}$	S4 done (same lock); S5 moot	blocked
4	ζ, signs $(--+)$	$(2+\mathrm{i}\sqrt{6})/4$	0.612	*inside* Ast ($s \approx -2.45$)	blocked (two-dimensional image, open mapping)	S4 done (same lock); S5 moot	blocked
5	$(-192303 + 85995\sqrt{5})/2$	$(3+\mathrm{i}\sqrt{15})/6$	0.646	inside Ast ($s \approx -6.17$)	blocked (two-dimensional image); also exterior	S4 done (`cert0_n5_e56.py` pair lock); S5 moot	blocked
9	$347648256 + 141926400\sqrt{6}$	$\mathrm{i}\sqrt{42}/42$	0.154	outside Ast ($c > 4$ real)	*exterior*: wrong sheet, deep below $1/\sqrt{2}$	S4 open (disc. -168, $h = 4$); S5 moot	blocked
10	$347648256 - 141926400\sqrt{6}$	$\mathrm{i}\sqrt{42}/14$	0.463	outside Ast ($s \approx 995$, $c > 4$ real)	*exterior*: wrong sheet below $1/\sqrt{2}$	S4 open; S5 moot	blocked

The conjectured right-hand sides are Samart's [2, Table 6]; entries #3 and #4 are the two remaining sign patterns of Theorem 1.6's family (the other two being Theorems 1.6 and 8.2). All five remaining entries are *blocked*, by two genuine obstructions analyzed in Section 10: the two-dimensional critical image (#4, #5: the $s = 81$ mechanism of [6]), and the wrong-sheet phenomenon of the EK series below $\mathrm{Im}\tau = 1/\sqrt{2}$ (#3, #9, #10, and again #5), where the conjectured identity concerns the true Mahler measure but the series side of the machine has no valid input. The five entries that this scoring marked as within current reach — #1, #2, #6, #7, #8 — are now proved as Theorems 8.1–8.5 (Section 8).

Beyond Table 6, three further open directions are discussed in Section 10: Samart's "three modular L-value" hypothesis $s = 16 + 1600\sqrt[3]{2} - 1280\sqrt[3]{4}$ [2], which has no attached CM point in the s_4-image; the exterior entries, for which a corrected formula of Samart–Tao $\widetilde{n}$-type [7] would be needed; and the Boyd-lineage two-variable conjectures, where no CM evaluation is available at all.

2 The machine

This section axiomatizes the method of [5, §3–§5] in family-independent language. The running example is the n_4-family, but the only family-specific inputs are the parametrization, the critical image, and the anchor identity; we display them as separate *stations* S1–S5, matching the checklist of §1.3.

2.1 Families, parametrizations, critical images

Definition 2.1 (Tempered family)**.** A *tempered family* is a Laurent-polynomial family $P_c(x_1,\dots,x_n) = P(x_1,\dots,x_n)+c$, $c\in\mathbb{C}$, such that $c\mapsto \mathrm{m}(P_c)$ is continuous on all of $\mathbb{C}$ — for the families here this holds by the elementary dominated-convergence argument of [5, Lemma 2.1]. A *critical image* is any closed set $K\subset\mathbb{C}$ containing the full image $P(\mathbb{T}^n)$; on $\mathbb{C}\setminus K$ the polynomial P_c does not vanish on $\mathbb{T}^n$, the integrand $\log|P_c|$ is real-analytic in c, and $c\mapsto\mathrm{m}(P_c)$ is real-analytic. (Taking only the critical *values* of P would not suffice: P_c vanishes on $\mathbb{T}^n$ for every $-c\in P(\mathbb{T}^n)$, including regular values — e.g. $P=x+x^{-1}$ on $\mathbb{T}^1$ has critical values ±2 but image $[-2,2]$. In both running examples below $K=\overline{P(\mathbb{T}^n)}$, the closure of the full image.)

Two families serve as running examples.

Example 2.2 (The n_2 family [5])**.** $P+c$ with $P=(x+x^{-1})(y+y^{-1})(z+z^{-1})$ and $c=\sqrt{s_2(\tau)}$, $s_2(\tau)=q^{-1}\prod_{n\ge1}(1+q^{2n-1})^{24}$. The critical image in the c-plane is the slit $[-8,8]$, and in the s-plane the segment $[0,64]$: one-dimensional.

Example 2.3 (The n_4 family)**.** $P_c=(x^4+y^4+z^4+1)/(xyz)+c$ after clearing the monomial (Mahler measure is unchanged), with the natural *holomorphic* parametrization

$$c(\tau)=\Big(\frac{\eta(2\tau)}{\eta(\tau)}\Big)^6\big(16A(\tau)^4+A(\tau)^{-4}\big),\qquad A(\tau)=\frac{\eta(\tau)\eta(4\tau)^2}{\eta(2\tau)^3},\qquad s_4(\tau)=c(\tau)^4. \tag{3}$$

Both c and s_4 are holomorphic on $\mathbb{H}$ — no branch choice is needed, in contrast to Example 2.2 — and s_4 has an integral q-expansion $s_4=q^{-1}\big(1+104q+4372q^2+\cdots\big)\in\mathbb{Z}((q))$ (exact integer check to order q^{30} in `cert0_s4_m144.py`). The critical image is *two-dimensional*: $P(\mathbb{T}^3)$ is the closed astroid disc

$$\mathrm{Ast}=\big\{c\in\mathbb{C}:\ |\mathrm{Re}c|^{2/3}+|\mathrm{Im}c|^{2/3}\le 4^{2/3}\big\}, \tag{4}$$

since P is a sum of four unit phasors with product 1 [6]. The branch cut of the holomorphic Mahler measure is the cross $\{c: c^4\in(0,256]\}=\big([-4,4]\cup\mathrm{i}[-4,4]\big)\setminus\{0\}$, contained in $\overline{\mathrm{Ast}}$; hence avoiding Ast automatically avoids every bad set, and we take $K=\mathrm{Ast}$.

Remark 2.4. The dimension of the critical image is the single most important topological invariant of a family for this method. A one-dimensional slit does not separate the plane: every point of the slit lies in the closure of the good domain, and the propagation lemma below reaches boundary points by continuity (this is what makes the $k=1$ proof of [5] possible). The two-dimensional astroid disc has interior: a CM parameter landing inside it (#4 and #5 of Table 1; the refuted $s=81$) lies inside an open bad region, no path can approach it from the good domain (open mapping), and the machine stops. See Section 10.

Definition 2.5 (The good domain and the EK expression)**.** For a holomorphic parametrization $c(\tau)$ set

$$V:=\big\{\tau\in\mathbb{H}:\ c(\tau)\notin K\big\}\qquad\text{(open, since } c \text{ is holomorphic)},$$

and let $F(\tau):=\mathrm{m}\big(P_{c(\tau)}\big)$ (times the family's normalization factor), continuous on all of $\mathbb{H}$ and real-analytic on V. An *EK datum* is a holomorphic function $E(\tau)$ on $\mathbb{H}$ with $G:=\mathrm{Re}E$ (in alternative normalizations $2\mathrm{Re}E$ or $\mathrm{Im}E$; all that matters is that G be the real part of a holomorphic function) real-analytic everywhere, such that $F=G$ on a nonempty open *anchor* set $U\subset V$. For the n_4 family (Example 2.3), with U_j as in (6) below,

$$E_4(\tau):=-\mathrm{i}\Big(2\pi\tau+\frac{10}{\pi^3}\big(U_1(\tau)-2U_2(\tau)\big)\Big),\qquad \mathrm{EK}_4(\tau):=\mathrm{Re}E_4(\tau)=\mathrm{Im}\Big[2\pi\tau+\frac{10}{\pi^3}\big(U_1-2U_2\big)\Big], \tag{5}$$

holomorphic, resp. real-analytic, on all of $\mathbb{H}$; the double series

$$U_j(\tau) := \sum_{m\neq 0} \frac{1}{m} \sum_{n\in\mathbb{Z}} (jm\tau + n)^{-3} \tag{6}$$

converges absolutely, locally uniformly on $\mathbb{H}$ [5, §2.4]. The lattice-sum form used for all CM evaluations is

$$\mathrm{EK}_4(\tau) = \frac{10\,\mathrm{Im}\tau}{\pi^3}\big(-T_1(\tau) + 4T_2(\tau)\big), \qquad T_d(\tau) := \sum_{\lambda\in\Lambda_d(\tau)}{}' \Big[\frac{2\mathrm{Re}\bar{\lambda}^2}{|\lambda|^6} + \frac{1}{|\lambda|^4}\Big], \tag{7}$$

with $\Lambda_d(\tau) := \mathbb{Z} + \mathbb{Z}\,d\tau$; the series is absolutely convergent, and (7) follows from $w^{-3} - \bar{w}^{-3} = -2\mathrm{i}(3u^2v - v^3)|w|^{-6}$ and $3u^2 - v^2 = 4u^2 - |w|^2$ exactly as in [5, Lemma 2.6].

2.2 The Mahler differential is single-valued

Proposition 2.6 (Mahler differential formula)**.** *Let P_c be a tempered family and $K \subset \mathbb{C}$ its critical image. Then*

$$\Omega(c) := \int_{\mathbb{T}^n} \frac{\mathrm{d}\mu}{P_c}$$

is single-valued and holomorphic on $\mathbb{C} \setminus K$, and as real 1-forms there,

$$\mathrm{dm}(P_c) = \mathrm{Re}\big[\Omega(c)\,\mathrm{d}c\big]. \tag{8}$$

Proof. For any compact $K_0 \Subset \mathbb{C} \setminus K$ one has $|P_c| \geq \mathrm{dist}(K_0, K) > 0$ uniformly for $c \in K_0$ on $\mathbb{T}^n$, so differentiation under the integral sign is legitimate and gives $\Omega'(c) = -\int_{\mathbb{T}^n} \mathrm{d}\mu/P_c^2$; holomorphy and single-valuedness follow (the integral is defined over the *fixed* cycle $\mathbb{T}^n$). Writing $c = c_1 + \mathrm{i}c_2$ and differentiating $\mathrm{m}(P_c) = \int \log|P_c|\mathrm{d}\mu$ under the integral sign gives $\partial_{c_1}\mathrm{m} = \int \mathrm{Re}(1/P_c)\mathrm{d}\mu = \mathrm{Re}\Omega$ and $\partial_{c_2}\mathrm{m} = \int \mathrm{Re}(\mathrm{i}/P_c)\mathrm{d}\mu = -\mathrm{Im}\Omega$, hence $\mathrm{dm} = \mathrm{Re}\Omega\,\mathrm{d}c_1 - \mathrm{Im}\Omega\,\mathrm{d}c_2 = \mathrm{Re}[\Omega\,\mathrm{d}c]$. The sign is $+$ under the present $P + c$ convention (it is $-$ in the $f - c$ convention of [5, Lemma 2.7]). □

Remark 2.7. Proposition 2.6 is the point at which all monodromy and universal-cover arguments disappear from the method. The holomorphic Mahler measure $\widetilde{\mathrm{m}}$ of Villegas [9] is multivalued, but its multivaluedness lives in locally constant imaginary periods: every branch satisfies $\widetilde{\mathrm{m}}'(c) = \Omega(c)$ on $\mathbb{C} \setminus K$. Comparing *differentials* rather than functions therefore never sees a branch choice, and the identity theorem below applies to a genuine single-valued holomorphic function.

2.3 The propagation theorem

Lemma 2.8 (Propagation lemma)**.** *Let $U \subset \mathbb{C}$ be connected open, $Z \subset U$ relatively closed, and $H : U \to \mathbb{R}$ continuous and locally constant on $U \setminus Z$. Then H is constant on every connected component W of $U \setminus Z$; if $H|_W \equiv C$ then $H|_{\overline{W}\cap U} \equiv C$.*

Proof. A locally constant function is constant on each connected component. For the second assertion, let $p \in \overline{W} \cap U$ and choose $p_n \in W$ with $p_n \to p$; continuity gives $H(p) = \lim_n H(p_n) = C$. □

Remark 2.9. The conclusion cannot be upgraded to "H constant on U": the devil's staircase is continuous, locally constant off the Cantor set, yet non-constant. Membership of the target point in $\overline{W}$ is a genuine topological input — supplied in our applications either by Samart's proved region (Theorems A, C, D, E), or by an explicitly certified path (Theorems B, F), and *absent* for the blocked entries of Table 1.

Theorem 2.10 (Main theorem: differential-comparison continuation). *Let P_c be a tempered family with critical image K, let $c(\tau)$ be holomorphic on $\mathbb{H}$, set $V = \{\tau : c(\tau) \notin K\}$, and let $F(\tau) = \mathrm{m}(P_{c(\tau)})$ (with the family normalization). Let $E(\tau)$ be holomorphic on $\mathbb{H}$, set $G := \mathrm{Re}E$ (the n_4 normalization $G = \mathrm{EK}_4 = \mathrm{Re}E_4$ of (5); rescaling E absorbs any other convention), and suppose $F = G$ on a nonempty open subset of a connected component W of V. Then*

$$F(\tau) = G(\tau) \qquad \textit{for every } \tau \in \overline{W} \cap \mathbb{H}. \tag{9}$$

Proof. Write $F = \nu\, \mathrm{m}(P_{c(\tau)})$ with the family's constant normalization factor ν ($\nu = 4$ for the n_4 family). On V both $\Omega(c(\tau))c'(\tau)$ and $E'(\tau)$ are holomorphic, so $\Psi(\tau) := \nu\, \Omega(c(\tau))c'(\tau) - E'(\tau)$ is holomorphic on V. By the chain rule applied to (8), $\mathrm{d}F = \nu\, \mathrm{Re}[\Omega(c(\tau))c'(\tau)\, \mathrm{d}\tau]$ on V, and $\mathrm{d}G = \mathrm{Re}[E'(\tau)\, \mathrm{d}\tau]$ there; hence $\mathrm{d}(F - G) = \mathrm{Re}[\Psi(\tau)\, \mathrm{d}\tau]$. On the anchor open set $F - G = 0$, so $\mathrm{Re}[\Psi\, \mathrm{d}\tau] = 0$ in all directions $\mathrm{d}\tau$; testing $\mathrm{d}\tau = 1$ and $\mathrm{d}\tau = \mathrm{i}$ gives $\Psi = 0$ there, and the identity theorem on the connected open set W gives $\Psi \equiv 0$ on W. Thus $H := F - G$ is locally constant on W, constant since W is connected, and zero because it vanishes on the anchor; H is continuous on all of $\mathbb{H}$ (Mahler continuity, Definition 2.1; G is real-analytic everywhere), so Lemma 2.8 with $U = \mathbb{H}$, $Z = c^{-1}(K)$ gives $H \equiv 0$ on $\overline{W} \cap \mathbb{H}$. ☐

Remark 2.11 (The five stations). Instantiating Theorem 2.10 for a conjectured identity $n_4(s) = R$ at a CM parameter splits into the five stations of §1.3: S1 identifies τ_0 with $s_4(\tau_0) = s$; S2 locates $c(\tau_0)$ relative to Ast; S3 certifies $\tau_0 \in \overline{W}$ (or gets it for free from Samart's proved region); S4 upgrades the numerical coincidence $s_4(\tau_0) = s$ to an exact algebraic lock (§2.5.3); S5 evaluates $\mathrm{EK}_4(\tau_0)$ exactly as a rational linear combination of L-values (§2.5.4). Theorem 2.10 then yields $n_4(s_4(\tau_0)) = \mathrm{EK}_4(\tau_0)$, and S4+S5 turn this into the identity $n_4(s) = R$.

2.4 The anchor identity for the n_4 family

The anchor input is Samart's theorem, quoted with one repair.

Theorem 2.12 (Samart, [1, Prop. 2.1(iii)]). *For pure imaginary $\tau = \mathrm{i}y$ with $y \geq 1/\sqrt{2}$, $n_4\big(s_4(\tau)\big) = \mathrm{EK}_4(\tau)$.*

Lemma 2.13 (Composite continuation lemma: the anchor on $|s_4| > 256$). *Let $D \subset \mathbb{H}$ be open with $|s_4(\tau)| > 256$ for all $\tau \in D$. Then $n_4(s_4(\tau)) = \mathrm{EK}_4(\tau)$ on D.*

Proof sketch with quoted inputs. Three quoted ingredients compose. (i) *Rogers' evaluation* (Prop. 2.2 and Thm. 2.3 of [8]): the ${}_5F_4$ hypergeometric expression for n_4 converges for $|s| > 256$ ($\sum b - \sum a = 3/2 > 0$) and equals $n_4(s)$ there: both sides are real parts of holomorphic functions of s on $\mathbb{C} \setminus \mathrm{Ast}$, equal on the real ray $s > 256$ by Rogers' theorem, hence equal on the connected open set $\{|s| > 256\}$ by the identity theorem. (ii) *Rogers' transformation* [8, Thm. 2.3]: $f_4(s_4(q)) = -\frac{1}{3}G(q) + \frac{2}{3}G(q^2)$ is an identity of functions analytic near $q = 0$, extending to $|q| < 1$ by the identity theorem (the composition with (i) is then restricted to the connected component of $\{\tau : |s_4(\tau)| > 256\}$ containing $\mathrm{i}\infty$, which is where every application lies). (iii) *Samart's computation* [1, proof of Prop. 2.1(iii)]: the identification of the q-series in (ii) with the U-series expression EK_4 uses termwise integration and Fourier expansions whose convergence requires only $|q| < 1$. The single step of Samart's proof that genuinely needs real q — the comparison $|s_4(q)| \geq s_4(\mathrm{e}^{-\pi\sqrt{2}}) = 256$ by real-variable monotonicity, which locates his argument in the convergence region — is replaced here by the hypothesis $|s_4| > 256$ on D, certified by interval arithmetic in each application (Theorems 1.3 and 1.7). This repair was identified by a line-by-line check of the proof of [1, Prop. 2.1(i)] and of the convergence analysis of [8, Thm. 2.3] (Samart's proof of Prop. 2.1(iii) itself is one sentence, referring to the (i)-proof); no other step uses the pure-imaginary hypothesis. ☐

Lemma 2.13 upgrades Samart's theorem from a statement on a half-line to an anchor on an open set: the anchor box $D = \{|\mathrm{Re}\tau| \le 1/32,\ |\mathrm{Im}\tau - 1| \le 1/32\}$ carries the certified bound $|s_4| \ge 493$ (Section 4), so $F = G$ on D, and Theorem 2.10 propagates the identity through the whole component W of $V_4 := \{\tau : c(\tau) \notin \mathrm{Ast}\}$ containing D. For the deep pure-imaginary points of Theorems A, C, D, E even this is unnecessary: Theorem 2.12 applies directly.

2.5 The certifier: `cert0`, `cert2`, `verify_P1`

The non-elementary computations are certified by interval arithmetic. The object entering the proofs is not a program but a *certificate*: finite data (parameter blocks with strict enclosures, or exact rational equalities) whose validity is checked by deterministic, finite, independently re-runnable computations. The trusted base consists of exactly three items: (i) the mathematical lemmas below (tail bounds, inclusion property); (ii) the documented outward-rounding semantics of the interval arithmetic employed (`mpmath.iv` rounds interval endpoints outward, with guard digits for the complex transcendental functions [33, 34]); and (iii) the public source code of the scripts. Item (ii) is replaceable: any outward-rounded interval implementation re-executing the same block lists obtains the same verdicts. All scripts are plain Python 3.12 + `mpmath` [33] (standard library `fractions` for the exact algebra), run independently of one another.

2.5.1 The inclusion property and tail bounds

Lemma 2.14 (Inclusion property, [35]). *Let E be any expression built from interval-enclosed constants, the variable, interval arithmetic operations, and interval elementary functions, all with outward-rounded semantics. Then for every interval box I in its domain, $E(I)$ contains the true value $E(\tau)$ for every $\tau \in I$.*

Proof. Induction on the structure of E; see [5, Lemma 6.1]. □

Lemma 2.15 (Tail bound for eta-quotient products). *Let $r := \overline{|q|} < 1$ be a certified upper bound. For $d \ge 1$, $e \in \mathbb{Z}$ and the principal branch of* log*,*

$$\Big|\log \prod_{n>N} \big(1 - q^{dn}\big)^{e}\Big| \le E := \frac{|e|\, r^{d(N+1)}}{(1 - r^d)(1 - r)},$$

so the truncated product carries the rigorous tail factor $\exp(z)$ *with z enclosed in the box $\{u + \mathrm{i}v : |u|, |v| \le E\}$ (evaluated as the interval exponential* of that box*).*

Proof. For $|z| \le r < 1$, $|\log(1 - z)| \le |z|/(1 - r)$; sum the geometric series, cf. [5, Lemma 6.2]. Products and quotients of such factors (the parametrization (3) is a quotient of two products) are enclosed numerator- and denominator-wise before division. □

On the certified paths $r \le \mathrm{e}^{-2\pi \cdot 0.702} < 0.0122$, and with truncation $N = 40$ the tail width satisfies $E < 10^{-78}$, far below the working precision.

2.5.2 `cert2`: path certification

A *path certificate* for a path $\gamma \subset \mathbb{H}$ is a finite list of parameter blocks I_ν covering γ (down to the endpoint, if the endpoint lies in V), with complex intervals Z_ν such that $c(I_\nu) \subset Z_\nu$ and every Z_ν is certified disjoint from Ast — i.e. the interval lower bound of $|\mathrm{Re}z|^{2/3} + |\mathrm{Im}z|^{2/3}$ over Z_ν exceeds $4^{2/3}$, a pure interval endpoint comparison. By Lemma 2.14 and Lemma 2.15 this is a proof, not an estimate (cf. [5, Prop. 6.4]).

Remark 2.16 (Generation and termination of the adaptive search). Blocks are found by adaptive bisection: accept a block when its enclosure is certified outside Ast, else bisect. Termination is not part of the trusted base, but it is guaranteed: the natural interval extension of an expression built from Lipschitz non-singular operations on a box has width $O(\text{diam})$ [35]; the factors of (3) are non-singular ($|q| < 1$, $|1 \pm q^{dn}|$ bounded away from 0) and the tail width is $< 10^{-78}$ uniformly; and on each compact path piece the true image has positive distance to Ast. Once the enclosure width drops below that margin, the block is accepted. (For a path ending at a point of $c^{-1}(K)$ — which does *not* occur in this paper — the same machinery certifies down to a cap $(y_0, y_0 + \varepsilon_0]$, closed analytically by a first-order Taylor expansion with a Cauchy-estimate remainder and a certified enclosure of the derivative at the endpoint; see [5, §4.3]. Theorems 1.3 and 1.7 have their endpoints strictly inside V_4, so no cap is needed.)

2.5.3 `cert0`: exact algebraic locks

The exactness of $s_4(\tau_0) = s$ follows a four-step pattern: (i) *integrality*: $s_4 \in \mathbb{Z}((q))$ (exact integer check), so by Shimura's CM theorem [12] the value of the eta-quotient (a modular unit with integral q-expansion) at a CM point is an algebraic integer in the corresponding ring class field; the class number bounds the degree, and conjugates are located via the level structure; (ii) *symmetric functions*: interval locks with rigorous tails force the elementary symmetric functions of the conjugate values (e.g. sum S and product P for a quadratic value) to be explicit integers; (iii) *pinning*: an interval enclosure of the target with radius smaller than half the separation of the candidate roots singles out a unique root; (iv) *reality*: real q at the chosen points ($q(\mathrm{i}y) > 0$, $q((1 + \mathrm{i}y)/2) < 0$) makes the products real and simplifies the locks. No exact equality is ever inferred from a floating-point approximation alone: each lock is a discreteness argument over an explicit finite candidate set.

2.5.4 `verify_P1`: the three-track CM evaluation

The evaluation $\mathrm{EK}_4(\tau_0) = R$ (station S5) is certified by three independent tracks, any one of which is sufficient in principle:

- **[X]/[S] exact algebra.** The lattice decomposition of $-T_1 + 4T_2$ into Hecke and Dirichlet sums, the inclusion–exclusion coefficients, the closed forms of even-character values $L(\chi, 2)$ [16], the functional-equation constants, and the final coefficient comparison in a rational *e-basis* are all carried out in rational/algebraic-integer arithmetic (`fractions.Fraction`); no interval estimates enter. The anchors of this track are quoted standard inputs listed explicitly in each application section (Hecke's theorem on grössencharacter theta series [13], class numbers, Dirichlet functional equations, Shimura's CM theorem).

- **[V] interval locks.** The same identity is re-proved by outward-rounded interval arithmetic: Poisson row sums with coth / tanh closed rows and rigorous Fourier tails for the lattice sums, E_1-continued-fraction-bracketed Mellin integrals for the L-values, Euler–Maclaurin for the Dirichlet values. Achieved half-widths are quoted in each application section (down to 9.1×10^{-53}).

- **mp cross-checks.** Independent 60-digit (`mpmath`, non-interval) reimplementations of every quantity, agreeing typically to 10^{-60}; these corroborate the implementation but do not enter the proof.

Newform identifications use Ligozat's eta-quotient criteria [14] (weight, character, cusp orders), exact-integer theta identity checks to order q^{60} against the Sturm bound [15] (which is ≤ 6 in all our cases), and the LMFDB labels [32]. Level and root number of each twist are determined by the *Fricke ratio* $f\big(\mathrm{i}/(\sqrt{N}\,y)\big)/\big(y^3 f(\mathrm{i}y/\sqrt{N})\big)$ evaluated at two ordinates — see §5.4 for why the naive "functional-equation residue" test is vacuous.

2.5.5 Script inventory

script	content
`cert0_s4_s7pair.py`	exactness $s_4(\mathrm{i}\sqrt{7})$, $s_4(\mathrm{i}\sqrt{7}/2) = 8292456 \pm 3132675\sqrt{7}$ (Theorem A)
`verify_P1_n4_s7pair.py`	Theorem A: exact separation track [S1]–[S10], FE proofs (§3.3), interval locks [V0]–[V6]
`cert0_s4_m144.py`	exactness $s_4(\tau_1) = -144$ (Theorem B)
`cert2_path_n4_m144.py`	anchor box + certified path for Theorem B (template for Theorem F)
`verify_P1_n4_m144.py`	Theorem B: 32 checks, four tracks including [G] Sturm-level newform identification and [V] interval locks
`n4_m144_true_mahler.py`	true Mahler side of Theorem B by spectral torus integration (22-digit collision)
`cert0_n4_p3_t1t2.py`	exactness of the s_4-values of Theorems C, D and of the conjugate of C (Theorem F's s-value)
`verify_P1_n4_p3_t1t2.py`	Theorems C, D: three tracks, [G] identifications, Fricke-ratio level determinations
`cert0_n4_p4_t3.py`	Theorem E: four-point quartic lock of the s_4-value
`verify_P1_n4_p4_t3.py`	Theorem E: three tracks, genus-character decomposition, [V] locks
`n4_p4_t4_cert.py`	certified path for Theorem F (interior-point verdict, margin 0.11590)
`n4_p4_t4_verify.py`	Theorem F: three tracks, shifted coth / tanh row machine, [V] locks
`n5_line_cert.py`	certified vertical segment $\mathrm{Re}\tau = 1/2$, $0.702 \leq \mathrm{Im}\tau \leq \sqrt{21}/2$ (station S3 of Theorems G–K and Remark 1.1; 8 blocks, minimum margin 0.00729)
`cert0_n5_e56.py`	Theorem I: $\mathbb{Q}(\sqrt{5})$ pair lock, $X^2 + 192303X + 1185921$
`cert0_n5_e78.py`	Theorems J, K: $\mathbb{Q}(\sqrt{3})$ pair lock, $X^2 + 1787904X + 84934656$
`verify_P1_n5_e1.py`	Theorem G: 49 checks, three tracks, [V] locks
`verify_P1_n5_e2.py`	Theorem H: 55 checks, three tracks, genus-character decomposition, [V] locks
`verify_P1_n5_e6.py`	Theorem I: 44 checks, three tracks, conductor-2-order lattice decomposition, [V] locks
`verify_P1_n5_e78.py`	Theorems J, K: 73 checks, three tracks, four-class anchors, [V] locks

All computations use `mpmath` interval arithmetic at 40–70 decimal digits (`iv.dps`); the certification scripts print `PASS/FAIL` per check and terminate with an all-checks-passed line, re-verified by the author.

3 Application I: the $\sqrt{7}$ pair

Both CM points are pure imaginary and deep inside Samart's region:

$$\tau_A = \mathrm{i}\sqrt{7} \quad (\mathrm{Im}\tau_A = \sqrt{7} > 1/\sqrt{2}), \qquad \tau_B = \frac{\mathrm{i}\sqrt{7}}{2} \quad (\mathrm{Im}\tau_B = \tfrac{\sqrt{7}}{2} > 1/\sqrt{2}),$$

so Theorem 2.12 applies directly: station S3 is free. The work is S4 (the exact s_4-values) and S5 (the exact CM evaluations, *separately* for the two points — an earlier stage of this project had proved only the sum of the two identities; the ray-class decomposition below closes each point independently).

3.1 Station S4: the exact s_4-values

By `cert0_s4_s7pair.py` (four-step lock of §2.5.3): the two values are conjugate algebraic integers in $\mathbb{Q}(\sqrt{7})$ (Shimura integrality; $q(\tau_A) > 0$, $q(\tau_B) > 0$); interval locks force the symmetric functions to integers,

$$S = 16584912, \qquad P = 69257922561 = 3^{12} \cdot 19^4 \quad \big(\text{exact check: } 8292456^2 - 7 \cdot 3132675^2 = P\big),$$

and the root separation 1.6×10^7 dwarfs the lock radii $|s_4(\mathrm{i}\sqrt{7}) - r_+| \le 9.6\times 10^{-51}$, $|s_4(\mathrm{i}\sqrt{7}/2) - r_-| \le 3.6 \times 10^{-54}$. Hence exactly

$$s_4(\mathrm{i}\sqrt{7}) = 8292456 + 3132675\sqrt{7} =: r_+, \qquad s_4(\mathrm{i}\sqrt{7}/2) = 8292456 - 3132675\sqrt{7} =: r_-.$$

3.2 Station S5: the lattice decomposition

Write $\varpi = (1+\sqrt{-7})/2$, so $2\varpi = 1+\sqrt{-7}$, $K = \mathbb{Q}(\sqrt{-7})$, $h(-7) = 1$, $\mathcal{O}_K^\times = \{\pm 1\}$. The lattices of (7) at the two points are (exact set equalities, checks [S1]–[S2] of `verify_P1_n4_s7pair.py`):

- at τ_A: $\Lambda_1(\tau_A) = \mathbb{Z} + \mathrm{i}\sqrt{7}\mathbb{Z} = \mathcal{O}^{(2)}$ (the order of conductor 2, discriminant -28), and $\Lambda_2(\tau_A) = \mathcal{O}^{(4)}$ (conductor 4, discriminant -112);
- at τ_B: $\Lambda_2(\tau_B) = \mathcal{O}^{(2)}$, and $\Lambda_1(\tau_B) = \frac{1}{2}\Lambda'$, where $\Lambda' = 2\mathbb{Z} + \mathrm{i}\sqrt{7}\mathbb{Z}$ is the index-2 *non-ideal* sublattice of $\mathcal{O}^{(2)}$; by (-4)-homogeneity, $T_1(\tau_B) = 16\,T(\Lambda')$.

With $B(\Lambda) = \sum'_{\lambda\in\Lambda} |\lambda|^{-4}$, $G(\Lambda) = \sum' \bar{\lambda}^2 |\lambda|^{-6}$ (real: all three lattices are conjugation-stable), $T = 2G + B$, the ray-class decompositions

$$\begin{aligned} \mathcal{O}^{(2)} \setminus \{0\} &= \{\alpha \equiv 1 \ (2)\} \ \sqcup \ 2\mathcal{O}_K \setminus \{0\}, \\ \mathcal{O}^{(4)} &= 2\mathcal{O}^{(2)} \ \sqcup \ \{\alpha \equiv 1, 3 \ (4)\}, \\ \Lambda' &= 2\mathcal{O}^{(2)} \ \sqcup \ \{\alpha \equiv 1{+}2\varpi,\ 3{+}2\varpi \ (4)\}, \end{aligned}$$

the inclusion–exclusion coefficients 9/16 and 23/16 at the split prime $(2) = (\varpi)(\bar{\varpi})$ (exact, using $\varpi^2 + \bar{\varpi}^2 = -3$), and the mod-4 ray-class pairing of the character $\chi_{-4} \circ N$ (values $+1, +1, -1, -1$ on the four unit classes, check [X4]) give, all in exact rational arithmetic (checks [S3]–[S6]):

lattice	B	G	$T = 2G + B$
$\mathcal{O}^{(2)}$	$\frac{5}{4}\zeta_K(2)$	$3L_3$	$6L_3 + \frac{5}{4}\zeta_K(2)$
$\mathcal{O}^{(4)}$	$\frac{41}{64}\zeta_K(2) + z_V$	$\frac{13}{8}L_3 + L_3^{\mathrm{tw}}$	$\frac{13}{4}L_3 + 2L_3^{\mathrm{tw}} + \frac{41}{64}\zeta_K(2) + z_V$
Λ'	$\frac{41}{64}\zeta_K(2) - z_V$	$\frac{13}{8}L_3 - L_3^{\mathrm{tw}}$	$\frac{13}{4}L_3 - 2L_3^{\mathrm{tw}} + \frac{41}{64}\zeta_K(2) - z_V$

where $L_3 := L(g_7, 3)$, $L_3^{\mathrm{tw}} := L(g_7 \otimes \chi_{-4}, 3)$, and $z_V := L(\chi_{-4}, 2)L(\chi_{-28}, 2)$. The pointwise assemblies (check [S8], including the separation witness combA $\neq$ combB):

$$\begin{aligned} \mathrm{combA} &:= -T\big(\mathcal{O}^{(2)}\big) + 4T\big(\mathcal{O}^{(4)}\big) = 7L_3 + 8L_3^{\mathrm{tw}} + \frac{21}{16}\zeta_K(2) + 4z_V, \\ \mathrm{combB} &:= -16\,T(\Lambda') + 4T\big(\mathcal{O}^{(2)}\big) = -28L_3 + 32L_3^{\mathrm{tw}} - \frac{21}{4}\zeta_K(2) + 16z_V. \end{aligned}$$

3.3 The functional equations and root numbers

Two functional-equation inputs are proved in `verify_P1_n4_s7pair.py`, not quoted: $M_7 = \frac{7\sqrt{7}}{4\pi^3} L_3$ [5, Lemma 2.8] and the twisted identity

$$U_{a/4}\, W_{112} = 4\,\gamma_a\, W_7\, U_{a/4} \qquad (a = 1, 3), \qquad \gamma_a = \begin{pmatrix} (1+7a^2)/4 & a \\ 7a & 4 \end{pmatrix} \in \Gamma_0(7), \tag{10}$$

an exact matrix identity (check [X10], with $\chi_{-7}(4) = +1$), which gives $h|[W_{112}]_3 = \mathrm{i}\, h$ for $h = g_7 \otimes \chi_{-4}$ — the Atkin–Li twist-W technique of [17] made self-contained for these parameters — hence root number $+1$ at level 112 and $M_7^{\mathrm{tw}} = \frac{112\sqrt{7}}{\pi^3} L_3^{\mathrm{tw}}$. The remaining quoted constants are the Dirichlet functional equations (textbook) and the finite-sum evaluation

$$L(\chi_{-28}, 2) = \frac{2\pi^2}{7\sqrt{7}},$$

whose character sum $\sum_a \chi_{-28}(a) a^2 = 448$ is an exact integer check [X7] [16].

3.4 Assembly

With the e-basis $e = \frac{\sqrt{7}}{\pi^3}\big(L_3, L_3^{\mathrm{tw}}, \zeta_K(2), z_V\big)$ one has $M_7 = \frac{7}{4} e_1$, $M_7^{\mathrm{tw}} = 112 e_2$, $d_7 = \frac{21}{2} e_3$, $d_4 = 7 e_4$ (exact, [S9]); the coefficient comparison

$$\mathrm{EK}_4(\mathrm{i}\sqrt{7}) = 10\,\mathrm{combA} \cdot e = \Big(70, 80, \tfrac{105}{8}, 40\Big) \cdot e = \frac{5}{28}\big(4M_7^{\mathrm{tw}} + 224 M_7 + 32 d_4 + 7 d_7\big),$$

$$\mathrm{EK}_4\Big(\frac{\mathrm{i}\sqrt{7}}{2}\Big) = 5\,\mathrm{combB} \cdot e = \Big(-140, 160, -\tfrac{105}{4}, 80\Big) \cdot e = \frac{5}{14}\big(4M_7^{\mathrm{tw}} - 224 M_7 + 32 d_4 - 7 d_7\big),$$

is exact rational arithmetic (check [S10]), and the two right-hand sides are distinct — the separation is complete and each identity holds individually. Combined with Theorem 2.12 and §2.5.3 this proves Theorem 1.2. □

Remark 3.1 (Independent confirmations). The mp track (60 dps, independent implementations: Mellin integrals, Poisson row sums, ray-class sums) confirms every step to 9×10^{-60}; the interval-lock track [V0]–[V6] (`iv.dps`$= 70$: continued-fraction bracketing, Fourier tail bounds, Euler–Maclaurin) locks the two final identities with half-widths 1.15×10^{-52} and 2.30×10^{-52}. Reference values:

$$M_7 = 0.1026716077789020112104565948982929139989\ldots,$$
$$M_7^{\mathrm{tw}} = 9.887687024790914246588215159482724883665\ldots$$

4 Application II: $n_4(-144)$

The CM point

$$\tau_1 = \frac{1+\sqrt{-3}}{2}$$

is *not* pure imaginary, so Theorem 2.12 does not apply directly; this is the first use of the full continuation machine in the n_4 family. It also shows the machine in its shortest form: the target is a strict interior point of V_4, so no endpoint cap and no boundary-continuity step are needed.

4.1 Station S3: the certified path

The path γ from the anchor box $D = \{|\mathrm{Re}\tau| \le 1/32,\ |\mathrm{Im}\tau - 1| \le 1/32\}$ to τ_1 — horizontal at $\mathrm{Im}\tau = 1$ to $1/2+\mathrm{i}$, then vertical at $\mathrm{Re}\tau = 1/2$ down to τ_1 — is certified inside V_4 by `cert2_path_n4_m144.py` (machinery of §2.5.2): the anchor box passes as a single block with astroid margin 0.291 and the certified bound $|s_4| \ge 493 > 256$ (which activates Lemma 2.13: $F = G$ on D); the horizontal leg takes 2 blocks (minimum margin 0.148); the vertical leg takes 1 block (margin 1.051); and τ_1 itself satisfies $c(\tau_1) = 2\sqrt{3}\,\mathrm{e}^{-\mathrm{i}\pi/4}$, whose astroid functional is $3.634 > 4^{2/3} = 2.520$ — a strict exterior point of Ast, so $\tau_1 \in V_4$ itself and $\tau_1 \in W$. The endpoint Taylor cap of [5] is absent. By Theorem 2.10, $n_4(s_4(\tau_1)) = \mathrm{EK}_4(\tau_1)$.

4.2 Station S4: the exact s_4-value

`cert0_s4_m144.py`: $q(\tau_1) = -\mathrm{e}^{-\pi\sqrt{3}}$ is a negative real, all products are real, and the value is pinned to a *rational integer* (Shimura integrality plus $h(-3) = 1$, so the ring class field is K; reality from the integral q-expansion and T-invariance): $|s_4(\tau_1)+144| \le 2.3\times10^{-51} < 1/2$, hence $s_4(\tau_1) = -144$ exactly.

4.3 Station S5: the CM evaluation

$K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega = (-1+\sqrt{-3})/2$, units μ_6. The structural simplification is the μ_6 *annihilation* $\sum_{u\in\mu_6} u^2 = 0$ (exact check [X1]): it forces $G(\mathcal{O}_K) = 0$, which is also the reason $\mathbb{Q}(\sqrt{-3})$ has no conductor-(1) grössencharacter of type $(2,0)$ and the evaluation involves a non-trivial conductor. The lattices are $\Lambda_1(\tau_1) = \mathcal{O}_K$ (since $\tau_1 = 1+\omega = -\omega^2$) and $\Lambda_2(\tau_1) = \mathcal{O}_2$ (conductor 2; 2 is inert), giving (exact [X2]–[X8]):

$$T_1(\tau_1) = 6\zeta_K(2), \qquad T_2(\tau_1) = 4L(g_{12},3) + \frac{9}{4}\zeta_K(2), \qquad -T_1 + 4T_2 = 16\,L(g_{12},3) + 3\zeta_K(2);$$

note the $\zeta_K(2)$-term does *not* cancel (coefficient 3) — this is the source of the d_3-term. The theta identity $2g_{12} = \sum'_{\alpha\equiv1\,(2)} \alpha^2 q^{N\alpha}$ is checked with exact integer arithmetic to order q^{60} (check [L1]); the newform identification $g_{12} = \eta(2\tau)^3\eta(6\tau)^3 \in S_3(\Gamma_0(12),\chi_{-3})$ is certified at Sturm level (checks [G1]–[G3]): Ligozat's criteria [14] (both congruences $24 \equiv 0 \bmod 24$; character $-2^3 3^3$, square kernel -3), all six cusp orders equal 1 (cuspidal; divisor degree $6 = \frac{3}{12}\cdot 24$), Sturm bound $\lfloor 3\cdot24/12\rfloor = 6 < 60$ [15]; LMFDB label `12.3.c.a` [32]. With $M_{12} = \frac{6\sqrt{3}}{\pi^3}L(g_{12},3)$ (root number $+1$, Fricke ratio) and $d_3 = \frac{3\sqrt{3}}{4\pi}L(\chi_{-3},2)$, the e-basis assembly gives (exact, [S1]):

$$\mathrm{EK}_4(\tau_1) = \frac{5\sqrt{3}}{\pi^3}\big(16L(g_{12},3) + 3\zeta(2)L(\chi_{-3},2)\big) = \frac{10}{3}\big(4M_{12} + d_3\big),$$

both sides being (80, 5/2) in the basis $\big(\sqrt{3}L(g_{12},3)/\pi^3,\ \sqrt{3}L(\chi_{-3},2)/\pi\big)$. Theorem 1.3 follows. $\square$

Remark 4.1 (Confirmations and the true-Mahler end). The mp track confirms the identity to 3.7×10^{-60} (check [E1]); the interval-lock track [V0]–[V3] (shifted coth / tanh rows, since $\mathrm{Re}\tau_1 = 1/2$) locks it with half-width 5.8×10^{-52}. Independently, direct spectral torus integration (`n4_m144_true_mahler.py`: $c(\tau_1)$ lies outside Ast, so the Jensen integrand is real-analytic periodic and the two-dimensional trapezoidal rule converges spectrally; residual 1.16×10^{-22} at $N = 128$) collides the two sides of Theorem 1.3 to 22 digits:

$$n_4(-144) = 5.09841965623737833323\ldots$$

The propagation argument does not depend on this numerical value. Reference value: $M_{12} = 0.30161498741294074646905293114776839988854\ldots.$

5 Application III: class number one, deep interior points (Theorems C and D)

Both CM points are pure imaginary and deep inside Samart's region,

$$\tau_C = \mathrm{i}\sqrt{2} \quad (\mathrm{Im}\tau_C = \sqrt{2}), \qquad \tau_D = 2\mathrm{i} \quad (\mathrm{Im}\tau_D = 2),$$

so Theorem 2.12 applies directly and the proofs consist of `cert0` locks and three-track CM evaluations. The new methodological content of this section is concentrated in station S5: the character $\chi_8 \circ N$ as an exact mod-2 ray-class projector, and the level determination of the twists, which contains a genuine trap.

5.1 The ray-class projector $\chi_8 \circ N$

In both fields the twist character materializes as a parity condition. Exact checks ([X4], [Y4] of `verify_P1_n4_p3_t1t2.py`):

- in $K = \mathbb{Q}(\sqrt{-2})$: for $a + b\sqrt{-2}$ of odd norm, $\chi_8\big(N(a + b\sqrt{-2})\big) = +1 \iff b$ even (a odd);
- in $K = \mathbb{Q}(\mathrm{i})$: for $a + b\mathrm{i}$ of odd norm, $\chi_8\big(N(a + b\mathrm{i})\big) = +1 \iff b \equiv 0\ (4)$ (a odd, b even).

Hence $(1 + \chi_8 \circ N)/2$ is exactly the projector onto the mod-2 ray class $\{\alpha \equiv 1\ (2)\}$, and in class number one no genus character is needed: every ideal decomposition in this section is a parity bookkeeping exercise certified in rational arithmetic. Also used everywhere below: the closed form

$$L(\chi_8, 2) = \frac{\pi^2\sqrt{2}}{16}$$

(even-character finite-sum formula [16]: $\tau(\chi_8) = 2\sqrt{2}$, $\sum_a \chi_8(a)a^2 = 16$, exact check [X7]).

5.2 Theorem C: $\tau_C = \mathrm{i}\sqrt{2}$

Station S4. `cert0_n4_p3_t1t2.py`: $q(\mathrm{i}\sqrt{2}) = \mathrm{e}^{-2\pi\sqrt{2}} > 0$ and $q\big((1 + \mathrm{i}\sqrt{2})/2\big) = -\mathrm{e}^{-\pi\sqrt{2}} < 0$ are real, so real interval locks suffice; the symmetric functions lock to the integers

$$S = 7312, \qquad P = -153664 = 3656^2 - 2 \cdot 2600^2$$

($|\Delta| \le 1.1 \times 10^{-49}$, 5.2×10^{-48}; root separation 7354), pinning

$$s_4(\mathrm{i}\sqrt{2}) = 3656 + 2600\sqrt{2}, \qquad s_4\Big(\frac{1 + \mathrm{i}\sqrt{2}}{2}\Big) = 3656 - 2600\sqrt{2}$$

— the second value is reused in Theorem 1.7.

Station S5. $K = \mathbb{Q}(\sqrt{-2})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-2}]$, $h = 1$, units ± 1, and $2 = -(\sqrt{-2})^2$ ramifies. The lattices are $\Lambda_1 = \mathcal{O}_K$ and $\Lambda_2 = \mathcal{O}_2 = \mathbb{Z} + 2\mathcal{O}_K$. With $L_8 := L(g_8, 3)$, $L_8^{\mathrm{tw}} := L(g_8 \otimes \chi_8, 3)$, and $z_V := L(\chi_8, 2)L(\chi_{-4}, 2)$, the inclusion–exclusion at the ramified prime (factors $1/4$ for B, $-1/4$ for G) gives (exact, [S]-track):

$$T(\mathcal{O}_K) = 4L_8 + 2\zeta_K(2), \qquad T(\mathcal{O}_2) = \frac{11}{4}L_8 + 2L_8^{\mathrm{tw}} + \frac{7}{8}\zeta_K(2) + z_V,$$

$$\mathrm{comb} := -T_1 + 4T_2 = 7L_8 + 8L_8^{\mathrm{tw}} + \frac{3}{2}\zeta_K(2) + 4z_V.$$

The newform identification $g_8 = \eta(\tau)^2\eta(2\tau)\eta(4\tau)\eta(8\tau)^2 \in S_3(\Gamma_0(8), \chi_{-8})$ is Sturm-level ([G]-track: Ligozat congruences $24 \equiv 0$ (24); cusp orders $(1, \frac{1}{2}, \frac{1}{2}, 1)$; weighted divisor degree $3 = \frac{3}{12} \cdot 12$; Sturm bound 3; theta identity $2g_8 = \sum'_{\alpha\in\mathcal{O}_K} \alpha^2 q^{N\alpha}$ exact to q^{60}). In the e-basis $e = \frac{\sqrt{2}}{\pi^3}\big(L_8, L_8^{\mathrm{tw}}, \zeta_K(2), z_V\big)$:

$$M_8 = 4e_1, \qquad M_8^{\mathrm{tw}} = 32e_2 \quad [\text{level } 32,\ w = +1,\ §5.4], \qquad d_8 = 24e_3, \qquad d_4 = 16e_4,$$

and the exact comparison ([S3]) $\mathrm{EK}_4(\mathrm{i}\sqrt{2}) = 10\,\mathrm{comb}\cdot e = (70, 80, 15, 40)\cdot e$ is the right-hand side of Theorem 1.4. $\square$

5.3 Theorem D: $\tau_D = 2\mathrm{i}$

Station S4. The same script locks

$$S = 286416, \qquad P = -126023688 = 143208^2 - 2\cdot 101574^2$$

($|\Delta| \le 4.1\times 10^{-48}$, 3.7×10^{-45}; separation 2.87×10^5), pinning $s_4(2\mathrm{i}) = 143208 + 101574\sqrt{2}$ and $s_4\big((1+2\mathrm{i})/2\big) = 143208 - 101574\sqrt{2}$ (the latter is proved as Theorem 8.1, Section 8).

Station S5. $K = \mathbb{Q}(\mathrm{i})$, $\mathcal{O}_K = \mathbb{Z}[\mathrm{i}]$, $h = 1$, units μ_4 — and $\sum_{u\in\mu_4} u^2 = 0$, so $G(\mathcal{O}_K) = 0$ (exact [Y1]; same annihilation mechanism as Theorem 1.3); $2 = -\mathrm{i}(1+\mathrm{i})^2$ ramifies. The lattices are $\Lambda_1 = \mathcal{O}_2$ and $\Lambda_2 = \mathcal{O}_4$. Every ideal coprime to 2 has exactly two generators $\equiv 1$ (2) ([Y3]), so the mod-2 ray sums carry *no twist term*: with $L_{16} := L(g_{16}, 3)$, $L_{16}^{\mathrm{tw}} := L(g_{16}\otimes\chi_8, 3)$, and $z_{V2} := L(\chi_8, 2)L(\chi_{-8}, 2)$,

$$T(\mathcal{O}_2) = 4L_{16} + \frac{7}{4}\zeta_K(2), \qquad T(\mathcal{O}_4) = \frac{9}{4}L_{16} + 2L_{16}^{\mathrm{tw}} + \frac{55}{64}\zeta_K(2) + z_{V2},$$

$$\mathrm{comb} = 5L_{16} + 8L_{16}^{\mathrm{tw}} + \frac{27}{16}\zeta_K(2) + 4z_{V2},$$

where the $\mathcal{O}_4$ decomposition uses $\mathcal{O}_4\setminus\{0\} = \{a \text{ odd},\ b\equiv 0\ (4)\} \sqcup 2\mathcal{O}_2\setminus\{0\}$ and the projector of §5.1 ([Y4]). The newform identification $g_{16} = \eta(4\tau)^6 \in S_3(\Gamma_0(16), \chi_{-4})$ is Sturm-level (five cusp orders all 1; weighted degree $6 = \frac{3}{12}\cdot 24$; Sturm bound 6; theta identity $2g_{16} = \sum_{\alpha\equiv 1\,(2)} \alpha^2 q^{N\alpha}$ of conductor (2) exact to q^{60}). In the e-basis $e = \frac{1}{\pi^3}\big(L_{16}, L_{16}^{\mathrm{tw}}, \zeta_K(2), z_{V2}\big)$:

$$M_{16} = 16e_1, \qquad M_{16}^{\mathrm{tw}} = 128e_2 \quad [\text{level } 64,\ w = +1], \qquad d_4 = 12e_3, \qquad d_8 = 64e_4,$$

and the exact comparison ([S6]) $\mathrm{EK}_4(2\mathrm{i}) = 20\,\mathrm{comb}\cdot e = (100, 160, 135/4, 80)\cdot e$ is the right-hand side of Theorem 1.5. $\square$

5.4 The twist-level trap and the Fricke ratio

Two methodological points emerged here and now belong to the machine.

(i) The "FE residue" test is vacuous. A widespread heuristic decides the root number of a completed L-function $\Lambda(f,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(f,s)$ by splitting the Mellin integral at $y = 1$ and comparing the two sides of $\Lambda(1) \stackrel{?}{=} w\Lambda(2)$. But the split gives $\Lambda(s) = I_1(s) + wI_2(3-s)$-type expressions for which, at $w = +1$, the two sides are *the same linear combination of the same two integrals at every candidate level N*: the "residue" vanishes identically in N and certifies nothing. (The same vacuity affected a check in the $\sqrt{7}$-pair script; there the conclusion was independently secured by the Fricke ratio, so no result was endangered.)

(ii) The non-vacuous discriminator is the Fricke ratio. For the true level N_0 one has the exact transformation $f\big(\mathrm{i}/(\sqrt{N_0}\,y)\big) = w\,y^3 f\big(\mathrm{i}y/\sqrt{N_0}\big)$ for *all* $y > 0$, coming from the Fricke involution (proved for each form in this paper either by the eta transformation law or by a twisted Fricke matrix identity such as (10)). Hence the ratio

$$R_N(y) := \frac{f\big(\mathrm{i}/(\sqrt{N}\,y)\big)}{y^3 f\big(\mathrm{i}y/\sqrt{N}\big)}$$

is identically w at $N = N_0$, while at $N \neq N_0$ it varies with y; evaluation at two ordinates separates the candidates. This is how the levels in this paper were fixed: 32 for $g_8 \otimes \chi_8$, 64 for $g_{16} \otimes \chi_8$ (two-ordinate agreement to 3.1×10^{-61} and better), and 96 for the twists of Section 6 (unique match in the candidate scan $\{24, 48, 96, 192, 384\}$).

(iii) The trap itself. For $g_{16} \otimes \chi_8$ the naive level guess $\mathrm{lcm}(16, 4^2)$-style reasoning suggests 32 — and the vacuous test (i) happily "confirms" it. With the wrong level-32 Mellin constant, M_{16}^{tw} comes out smaller by a factor $2\sqrt{2}$, and the conjectured identity fails numerically by $\sim 1\%$; a first pass over Theorem 1.5 recorded exactly this mismatch, briefly suggesting an $s = 81$-type refutation. The Fricke ratio at two ordinates gives the true level 64, and with it the identity holds to 2.5×10^{-60}. Since a wrong level *silently scales* an L-value by a simple algebraic factor, we record the rule: *never determine a twist level by a residual; determine it by the Fricke ratio at two ordinates, and cross-check against the assembled identity.*

Remark 5.1 (Confirmations for Theorems C and D). The mp tracks (60 dps) confirm the two final identities to 2.5×10^{-60} each; the interval-lock tracks [V] contain zero with half-widths $\leq 4.6\times10^{-53}$ (Theorem C) and $\leq 6.2\times10^{-53}$ (Theorem D). Direct torus integration ($c \approx 9.25$ and 23.14, smooth integrands) confirms both at float64 level. Reference values:

$$\begin{aligned}
M_8 &= 0.13531849542312691061500606545953958835 32\ldots, \\
M_8^{\mathrm{tw}} &= 1.5490848470956114862716581594443762817 85\ldots \quad (\text{level } 32,\ w = +1), \\
M_{16} &= 0.4975478545679851929276297713583412482363\ldots, \\
M_{16}^{\mathrm{tw}} &= 4.3363612492444456091350120197212782129 73\ldots \quad (\text{level } 64,\ w = +1), \\
d_8 &= \frac{4\sqrt{2}}{\pi} L(\chi_{-8}, 2) = 1.9171950931209540617988237536697845644585\ldots
\end{aligned}$$

(d_8 by the functional equation and by direct numerical differentiation, agreeing exactly at working precision).

6 Application IV: class number two (Theorem E)

The CM point $\tau_E = \mathrm{i}\sqrt{6}$ is pure imaginary with $\mathrm{Im}\tau_E = \sqrt{6} > 1/\sqrt{2}$, so Theorem 2.12 applies directly. The field $K = \mathbb{Q}(\sqrt{-6})$ has discriminant -24 and *class number* 2 — the first such case treated by the machine — and everything is one step richer: two newforms, a genus character, a non-principal ideal class, and a four-point `cert0` lock.

6.1 Station S4: the four-point quartic lock

Class number two makes the s_4-value quartic over $\mathbb{Q}$, in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. `cert0_n4_p4_t3.py` locates and locks all four conjugate values:

signs	CM point	lock half-width
$(+,+)$	$\mathrm{i}\sqrt{6}\ \ (q=\mathrm{e}^{-2\pi\sqrt{6}}>0)$	7.0×10^{-47}
$(-,+)$	$\mathrm{i}\sqrt{6}/4\ \ (q=\mathrm{e}^{-\pi\sqrt{6}/2}>0)$	4.4×10^{-51}
$(+,-)$	$(1+\mathrm{i}\sqrt{6})/2\ \ (q=-\mathrm{e}^{-\pi\sqrt{6}}<0)$	3.1×10^{-50}
$(-,-)$	$1/2+\mathrm{i}\sqrt{6}/4\ \ (q=-\mathrm{e}^{-\pi\sqrt{6}/2}<0)$	6.8×10^{-53}

with minimal polynomial

$$X^4-4829472X^3-8676927360X^2+3042408646656X+7520406736896,$$

whose coefficients are confirmed as integers both by exact $\mathbb{Z}[\sqrt{2},\sqrt{3}]$ arithmetic and by the interval locks; the minimum root separation 305.0 dwarfs every lock radius. Two structural observations aided the location (used for location only, not in the certified chain): s_4 is W_2-invariant, $s_4(-1/(2\tau))=s_4(\tau)$, observed numerically to 10^{-33} at sample points; and the second class of discriminant -24 degenerates: $s_4(\mathrm{i}\sqrt{6}/2)=s_4(\mathrm{i}\sqrt{6}/6)=2304\in\mathbb{Z}$, which is why that value is absent from the quartic roots. In particular the $(+,-)$ value is entry #2, proved as Theorem 8.2 (Section 8), and the $(-,\pm)$ values are entries #3, #4 of Table 1: their station S4 is already done.

6.2 The two newforms and the genus character

The class group is $\mathbb{Z}/2$ and coincides with the genus group. The theta series of the two grössencharacters ψ and $\psi\cdot\chi_g$ of type $(2,0)$ have coefficients (exact-integer checks [th0]–[th6]: integrality, $a(1)=1$, the Hecke recursion $a(p^2)=a(p)^2-\chi_{-24}(p)p^2$, multiplicativity, vanishing at inert primes, distinctness):

$$a^{(1)}(n)=P(n)-Q(n),\qquad a^{(2)}(n)=P(n)+Q(n),$$

$$P(n)=\frac{1}{2}\sum_{a^2+6b^2=n}{}'\big(a^2-6b^2\big),\qquad Q(n)=\frac{1}{4}\sum_{2x^2+3y^2=n}{}'\big(4x^2-6y^2\big),$$

P collecting the principal ideal class and Q the non-principal one (elements of norm $2n$ in $\mathfrak{p}_2=(2,\sqrt{-6})$ are in bijection with non-principal ideals of norm n). By Hecke's theorem [13] both are newforms of $S_3(\Gamma_0(24),\chi_{-24})$, which has dimension 2, so they exhaust the space; the coefficient match $(q\mp2q^2\pm3q^3+4q^4\pm2q^5-6q^6)$ fixes the LMFDB labels

$$g_{24}^{(1)}=P-Q=\mathsf{24.3.h.a},\qquad g_{24}^{(2)}=P+Q=\mathsf{24.3.h.b},$$

the unique rational CM newforms of level 24 and weight 3 [32] (in agreement with Schütt's classification [20]). Samart's $M_{24}^{(1)}$ (coefficient 28 in Theorem 1.6) is our $g_{24}^{(1)}=P-Q$; swapping the labels breaks the identity at the 0.4 level, a labeling pitfall we record. The *genus character* is $\chi_g=\chi_8\circ N=\chi_{-3}\circ N$ on ideal norms (exact check [th2]: $\chi_8\chi_{-3}=\chi_{-24}$, the field character, so the two compositions coincide on norms), and it handles exactly the non-principal class that the mod-2 ray projector of §5.1 cannot see.

6.3 Station S5: the class-number-two lattice decomposition

The lattices are $\Lambda_1=\mathcal{O}_K$ and $\Lambda_2=\mathcal{O}_2$. The new ingredients, relative to class number one, are: (a) the class-group projector $(1+\chi_g)/2$ on the B-side and the difference of the two ψ-values on the G-side; (b) sums over $\mathfrak{p}_2$ as a lattice, with homothety factors 1/4 (not 1/2 — numerically locked, then certified); (c) the ray condition $\{a \text{ odd}, b \text{ even}\}$ via $\chi_{-8}\circ N$, which is *not* a Hecke character (nontrivial on principal ideals) but whose L_K-value still factors as a product of Dirichlet L-functions. Exact decomposition:

$$B(\mathcal{O}_K) = \zeta_K(2) + L(\chi_8, 2)L(\chi_{-3}, 2), \qquad G(\mathcal{O}_K) = L\big(g_{24}^{(1)}, 3\big) + L\big(g_{24}^{(2)}, 3\big),$$
$$B(\mathfrak{p}_2) = \frac{\zeta_K(2) - L(\chi_8, 2)L(\chi_{-3}, 2)}{4}, \qquad G(\mathfrak{p}_2) = \frac{L(g_{24}^{(2)}, 3) - L(g_{24}^{(1)}, 3)}{4},$$

(the sign in $G(\mathfrak{p}_2)$ is $\psi(\mathfrak{p}_2) = -2$), and

$$B_{\mathrm{odd},\chi} = L(\chi_{-8}, 2)L(\chi_3, 2) + L(\chi_{-4}, 2)L(\chi_{24}, 2), \qquad G_{\mathrm{odd},\chi} = L\big(g_{24}^{(1)} \otimes \chi_{-8}, 3\big) + L\big(g_{24}^{(2)} \otimes \chi_{-8}, 3\big).$$

Note $T(\Lambda) = B(\Lambda) + 2G(\Lambda)$ for *every* lattice: the pointwise identity $4(\mathrm{Re}z)^2/|z|^6 - 1/|z|^4 = 2\mathrm{Re}(z^2)/|z|^6 + 1/|z|^4$ (exact check [X0]) plus absolute convergence, closing what had briefly appeared to be a gap. The even primitive closed forms [16]

$$L(\chi_3, 2) = \frac{\pi^2\sqrt{3}}{18} \quad \text{(conductor 12, } \textstyle\sum a^2\chi_3(a) = 48), \qquad L(\chi_{24}, 2) = \tfrac{\pi^2\sqrt{6}}{24} \quad (\textstyle\sum a^2\chi_{24}(a) = 288)$$

are absorbed by the d_8 and d_4 terms in the assembly. With $L_1, L_2, L_1^{\mathrm{tw}}, L_2^{\mathrm{tw}}$ the four L-values at $s = 3$,

$$\begin{aligned} \mathrm{comb} = {} & \frac{7}{2}L_1 + \frac{3}{2}L_2 + 4L_1^{\mathrm{tw}} + 4L_2^{\mathrm{tw}} + \frac{3}{4}\zeta_K(2) \\ & + \frac{7}{4}L(\chi_8, 2)L(\chi_{-3}, 2) + 2L(\chi_{-8}, 2)L(\chi_3, 2) + 2\,\mathrm{Cat}\,L(\chi_{24}, 2), \end{aligned}$$

where $\mathrm{Cat} = L(\chi_{-4}, 2)$ is Catalan's constant, and the [S1]–[S8] Fraction-exact assembly (single-term substitutions) produces the coefficient list

$$\tfrac{35}{12}, \; \tfrac{5}{4}, \; \tfrac{5}{12}, \; \tfrac{5}{12}, \; \tfrac{35}{12}, \; \tfrac{5}{2}, \; \tfrac{5}{6}, \; \tfrac{5}{48},$$

yielding Theorem 1.6. □

Remark 6.1 (Traps recorded for the machine). (i) χ_3 has conductor 12 with period values 0 at even positions; a wrongly tabulated period (nonzero even positions) makes standard Dirichlet-L software silently return a wrong value (1.1325 instead of 0.951 ...), which once produced a false failure at the 3.5×10^{-2} level. The machine now treats character tables as exact data to be certified ([X3]). (ii) The twist levels (96) were fixed by the Fricke-ratio scan of §5.4, never by residues.

Remark 6.2 (Confirmations). The mp track (60 dps) confirms the final identity to 2.49×10^{-60} (check [E]); the [V] interval-lock track (Dual-number coth rows, E_1-bracketed Mellin with the self-contained coefficient bound $|a(n)| \le 6n^3$, level-96 Mellin truncation $n_0 \approx 481$) locks $T(\mathcal{O}_K)$, $T(\mathcal{O}_2)$, and the final identity with half-widths 4.50×10^{-53}, 4.92×10^{-53}, and 9.72×10^{-53}. Direct torus integration ($c \approx 46.88$, smooth) confirms at float64 level. Reference values (50 dps, Mellin; level 96, $w = +1$ for the twists):

$$\begin{aligned} M_{24}^{(1)} &= 0.83669465148509836614171531082779903762 89\ldots, \\ M_{24}^{(2)} &= 1.09358062553574299562381618241798248742 3\ldots, \\ M_{24}^{(1),\mathrm{tw}} &= 8.54741283859595097226099097602224134889 7\ldots, \\ M_{24}^{(2),\mathrm{tw}} &= 7.18356961197733651253794048557790715997 9\ldots \end{aligned}$$

7 Application V: the boundary point that turned out interior (Theorem F)

The point

$$\tau_4 = \frac{1+\sqrt{-2}}{2}, \qquad \mathrm{Im}\tau_4 = \frac{1}{\sqrt{2}},$$

lies *exactly* on the boundary of Samart's applicability region. Moreover $q(\tau_4) = -e^{-\pi\sqrt{2}} < 0$ is a negative real (outside his $q \in (0,1)$ hypothesis) and $|s_4(\tau_4)| = 20.96\ldots < 256$ (outside the ${}_5F_4$ convergence region), so Theorem 2.12 fails to cover τ_4 for two independent reasons. This was expected to require a boundary-continuity argument. It does not: the boundary in question is not the boundary of the problem.

7.1 Samart's boundary is not the topological boundary of V_4

A 60-digit scan along the line $\mathrm{Re}\tau = 1/2$ shows the astroid functional $|\mathrm{Re}c|^{2/3} + |\mathrm{Im}c|^{2/3}$ of $c(\tau)$ decreasing monotonically from 4.377 at $y = 1$, crossing the threshold $4^{2/3} = 2.51984$ only at $y \approx 0.693$ (margin $+0.064$ at $y = 0.700$, -0.011 at $y = 0.690$). At τ_4 ($y = 1/\sqrt{2} = 0.70711$) the functional is 2.6357, with margin $+0.116$. Since c is holomorphic (an open map) and Ast is closed, τ_4 is a *strict interior point* of V_4. Samart's line $\{\mathrm{Im}\tau = 1/\sqrt{2}\}$ is dictated by the real-q mechanism of his proof; the topological boundary of V_4 along this segment lies ≈ 0.014 lower, and τ_4 sits strictly between the two. Consequently the short `cert2` path of Theorem 1.3 suffices, and no boundary-continuity step is needed.

7.2 Station S3: the certified path

`n4_p4_t4_cert.py` (template `cert2_path_n4_m144.py`; leg B extended down to $y = 0.702 < 1/\sqrt{2}$ so that τ_4 lies strictly inside the certified leg; the $|q|$ bound 0.0122 keeps the tail $r^{41} < 10^{-78}$):

- anchor box D: margin 0.29143, certified $|c| \geq 4.71$, $|s_4| \geq 493.6 > 256$ (activating Lemma 2.13);
- leg A ($y = 1$, $x \in [0, 1/2]$): 2 blocks, minimum margin 0.14759;
- leg B ($x = 1/2$, $y \in [0.702, 1]$): 4 blocks, minimum margin 0.02270 (most dangerous block $y \in [0.702, 0.7393]$);
- point enclosure at τ_4: astroid margin $0.11590 > 0$ — an interval-certified interior-point verdict;
- self-tests: 21/21 point inclusions (80-dps values inside interval enclosures, including τ_4 itself) and 5/5 enclosure-containment checks (point enclosures inside block enclosures).

Hence the whole path and τ_4 lie in W, the component of the anchor box, and Theorem 2.10 gives $n_4(s_4(\tau_4)) = \mathrm{EK}_4(\tau_4)$. As a byproduct, the entries #1 ($\tau = (1+2\mathrm{i})/2$, the top corner of the legs) and #7 ($\tau = (3+\mathrm{i}\sqrt{21})/6 = 1/2 + 0.7638\mathrm{i}$, on leg B) — now proved as Theorems 8.1 and 8.4 (Section 8) — already have their station S3 certified by this same path.

7.3 Station S4

The value $s_4(\tau_4) = 3656 - 2600\sqrt{2}$ is already locked exactly by `cert0_n4_p3_t1t2.py` as the conjugate point of Theorem 1.4 ($S = 7312$, $P = -153664$; §5.2).

7.4 Station S5: a different lattice decomposition, not a change of clothes

Although the s-value is conjugate to Theorem 1.4's, the CM point is different, and the lattice structure differs: $2\tau_4 = 1 + \sqrt{-2}$ gives $\Lambda_2 = \mathcal{O}_K$ (not $\mathcal{O}_2$), and

$$2\Lambda_1 = 2\mathbb{Z} + \mathbb{Z}(1+\sqrt{-2}) = \{a + b\sqrt{-2} : a \equiv b\ (2)\} =: \mathcal{O}' = 2\mathcal{O}_K \ \sqcup\ \big((1+\sqrt{-2}) + 2\mathcal{O}_K\big),$$

a non-ideal sublattice (exact [X1]–[X3]). The parity-class sums from Theorem 1.4's anchors ([X4], [X5], homothety 1/16 [X8]):

$$B_{00} = \frac{\zeta_K(2)}{8}, \quad B_{11} = \frac{3}{4}\zeta_K(2) - z_V, \qquad G_{00} = \frac{L_8}{8}, \quad G_{11} = \frac{5}{4}L_8 - L_8^{\mathrm{tw}},$$

$$T(\mathcal{O}') = \frac{11}{4}L_8 - 2L_8^{\mathrm{tw}} + \frac{7}{8}\zeta_K(2) - z_V, \qquad T(\Lambda_1) = 16\,T(\mathcal{O}'),$$

$$\mathrm{comb} = -T(\Lambda_1) + 4T(\Lambda_2) = -28L_8 + 32L_8^{\mathrm{tw}} - 6\zeta_K(2) + 16z_V.$$

Note the sign pattern: the twisted and $\zeta_K(2)$ terms flip sign relative to Theorem 1.4 (combinations $-28/+32$ versus $+7/+8$), and the $-d_8$ term traces to the coefficient -16 of z_V (exact witness [S7]: the two sign patterns are genuinely different). The e-basis assembly of §5.2 gives (exact [X10]/[S6])

$$\mathrm{EK}_4(\tau_4) = 5\,\mathrm{comb}\cdot e = (-140, 160, -30, 80)\cdot e = \frac{5}{4}\big(4M_8^{\mathrm{tw}} - 28M_8 + 4d_4 - d_8\big),$$

proving Theorem 1.7. □

Remark 7.1 (Confirmations). The mp track (60 dps; the L-values recomputed in-script at levels 8/32, cross-checked against Section 5 to 1.6×10^{-40}) confirms the final identity to 9.33×10^{-61} (check [E1]); the [V] interval-lock track — extended by a shifted coth / tanh row machine (`lattice_T_iv_t4`: at $x_0 = 1/2$ the Poisson kernel rows alternate between $\coth(\pi y)$ and $\tanh(\pi y)$; the power terms are shift-independent and the tail bounds uniform in the shift) — locks $T(\Lambda_2)$, $T(\Lambda_1)$, and the final identity with half-widths 3.48×10^{-53}, 6.94×10^{-52}, and 9.12×10^{-53}. Direct torus integration at the complex parameter $c = 1.5128932208(1-\mathrm{i})$ (outside Ast: functional 2.6357) confirms at float64 level: $n_4(3656 - 2600\sqrt{2}) \approx 3.5283920696 \approx \mathrm{EK}_4(\tau_4)$ (the two computations differ by 2.5×10^{-13}, the float64 noise floor).

8 Five further proofs: class numbers one, two, and four

The five entries of Samart's Table 6 that the checklist of §1.3 scored as within current reach are now proved. We state them as Theorems G–K and record the status of each station; the three genuinely new methodological ingredients are collected in §8.2.

Theorem 8.1 (Theorem G)**.** *With the notation of Theorem 1.5,*

$$n_4\big(143208 - 101574\sqrt{2}\big) = \frac{5}{8}\big(4M_{16}^{\mathrm{tw}} - 20M_{16} - 9d_4 + 4d_8\big).$$

The attached CM point is $\tau_0 = (1+2\mathrm{i})/2$.

Theorem 8.2 (Theorem H: class number two, second sign pattern)**.** *With the notation of Theorem 1.6,*

$$n_4\big(1207368 + 853632\sqrt{2} - 697680\sqrt{3} - 493272\sqrt{6}\big)$$
$$= \frac{5}{24}\Big(4M_{24}^{(1),\mathrm{tw}} + 4M_{24}^{(2),\mathrm{tw}} - 28M_{24}^{(1)} - 12M_{24}^{(2)} - 28d_3 + 24d_4 + 8d_8 - d_{24}\Big).$$

The attached CM point is $\tau_0 = (1+\sqrt{-6})/2$.

Theorem 8.3 (Theorem I: $\mathbb{Q}(\sqrt{-15})$)**.** *Let $K = \mathbb{Q}(\sqrt{-15})$ (class number 2), and let $g_{15}^{(1)}, g_{15}^{(2)}$ be the two newforms of $S_3(\Gamma_0(15), \chi_{-15})$, theta series of the two grössencharacters of K of type $(2,0)$ (LMFDB* 15.3.d.a *and* 15.3.d.b*, the only dimension-one CM forms of this level), numbered so that $a_2\big(g_{15}^{(1)}\big) = +1$ and $a_2\big(g_{15}^{(2)}\big) = -1$. With $M_{15}^{(i)} := L'\big(g_{15}^{(i)}, 0\big)$,*

$$n_4\Big(\frac{-192303 - 85995\sqrt{5}}{2}\Big) = \frac{1}{15}\big(120M_{15}^{(1)} + 160M_{15}^{(2)} + 88d_3 + 5d_{15}\big).$$

The attached CM point is $\tau_0 = (1+\sqrt{-15})/2$.

Theorem 8.4 (Theorem J: class number four)**.** *Let $K = \mathbb{Q}(\sqrt{-21})$ (discriminant -84, class number 4, class group $(\mathbb{Z}/2)^2$ with ideal classes $1, [\mathfrak{p}_2], [\mathfrak{p}_3], [\mathfrak{p}_6]$, $\mathfrak{p}_6 = \mathfrak{p}_2\mathfrak{p}_3$). The four-dimensional space $S_3(\Gamma_0(84), \chi_{-84})$ is spanned by the CM theta series*

$$g(\varepsilon_2, \varepsilon_3) = P_0 + \varepsilon_2 P_2 + \varepsilon_3 P_3 + \varepsilon_2\varepsilon_3 P_6, \qquad \varepsilon_2, \varepsilon_3 \in \{\pm 1\},$$

of the four grössencharacters of K of type $(2,0)$, where P_0, P_2, P_3, P_6 are the theta sums over the four ideal classes (LMFDB 84.3.d.**; the precise theta construction is documented in the header of* `verify_P1_n5_e78.py`*). With $M_{84}^{(3)} := L'\big(g(-1,+1), 0\big)$ and $M_{84}^{(4)} := L'\big(g(-1,-1), 0\big)$,*

$$n_4\big(-893952 + 516096\sqrt{3}\big) = \frac{20}{7}\big(M_{84}^{(3)} - M_{84}^{(4)} + 8d_3 - 4d_4\big).$$

The attached CM point is $\tau_0 = (3+\sqrt{-21})/6$.

Theorem 8.5 (Theorem K)**.** *With the notation of Theorem 8.4,*

$$n_4\big(-893952 - 516096\sqrt{3}\big) = \frac{20}{21}\big(M_{84}^{(3)} + M_{84}^{(4)} + 8d_3 + 4d_4\big).$$

The attached CM point is $\tau_0 = (1+\sqrt{-21})/2$.

8.1 Station status

Theorem G. S1/S2: $s_4(\tau_0) = 143208 - 101574\sqrt{2}$ ($|s| > 256$, outside Ast). S3 is free: $\tau_0 = (1/2, 1)$ is the top corner of the certified path of Section 7 (the intersection of legs A and B of `n4_p4_t4_cert.py`). S4 is quoted from `cert0_n4_p3_t1t2.py` (the conjugate lock of Theorem 1.5, §5.3). S5 is `verify_P1_n5_e1.py` (49 checks): $2\tau_0 = 1 + 2\mathrm{i}$ gives $\Lambda_2 = \mathcal{O}_2$, the conductor-2 order of $\mathbb{Q}(\mathrm{i})$ — the same lattice as Theorem 1.5's Λ_1 — and $2\Lambda_1 = 2\mathcal{O}_2 \sqcup \big((1+2\mathrm{i}) + 2\mathcal{O}_2\big)$; the exact track gives comb $= -20L_{16} + 32L_{16}^{\mathrm{tw}} - \frac{27}{4}z_C + 16z_{V_2}$ and $\mathrm{EK}_4(\tau_0) = (-200, 320, -135/2, 160)\cdot e$, the right-hand side of Theorem 8.1; the interval lock [V3] has half-width 1.23×10^{-52} and the mp track agrees to 1.87×10^{-60}. □

Theorem H. S3: $\tau_0 = (1+\mathrm{i}\sqrt{6})/2$ lies on the certified vertical segment of `n5_line_cert.py` (point-enclosure astroid margin 3.16; §8.2(i)). S4 is quoted from `cert0_n4_p4_t3.py` (the four-point quartic lock, point tag `'C'`; §6). S5 is `verify_P1_n5_e2.py` (55 checks): $2\tau_0 = 1+\sqrt{-6}$ gives $\Lambda_2 = \mathcal{O}_K$, and $2\Lambda_1 = \{c + d\sqrt{-6} : c \equiv d\ (2)\} = 2\mathcal{O}_K \sqcup C''$ with $C'' = \{c, d \text{ both odd}\}$ the $\chi_{-8}\circ N = -1$ side of the odd sums; with the anchors of §6.3 the exact track gives

$$\mathrm{comb} = -T(\Lambda_1) + 4T(\Lambda_2) = 3T(\mathcal{O}_K) - 16T(C'') = (-14, -6, 16, 16, -3, -7, 8, 8)$$

on the basis $\big(L_1, L_2, L_1^{\mathrm{tw}}, L_2^{\mathrm{tw}}, \zeta_K(2), L(\chi_8,2)L(\chi_{-3},2), L(\chi_{-8},2)L(\chi_3,2), \mathrm{Cat}\, L(\chi_{24},2)\big)$ of §6.3 — minus four times Theorem 1.6's combination with the character-dependent pieces sign-flipped, the same flip pattern as Theorems C/F — and $\mathrm{EK}_4(\tau_0) = (5\sqrt{6}/\pi^3)\,\mathrm{comb}$ ($\mathrm{Im}\tau_0$ halved relative to Theorem 1.6), the right-hand side of Theorem 8.2; [V3] half-width 1.94×10^{-52}, mp agreement 1.87×10^{-60}. □

Theorem I. S3: $\tau_0 = (1 + \mathrm{i}\sqrt{15})/2$ lies on the certified vertical segment (margin 9.54). S4 is `cert0_n5_e56.py`: a $\mathbb{Q}(\sqrt{5})$ pair lock of τ_0 together with the point $(3+\mathrm{i}\sqrt{15})/6$ of the blocked entry #5, minimal polynomial $X^2+192303X+1185921$ (discriminant $5\cdot 85995^2$), root separation 1.92×10^5 against lock widths $\le 2.8 \times 10^{-48}$. S5 is `verify_P1_n5_e6.py` (44 checks): $\tau_0 = \omega$ with $\mathcal{O}_K = \mathbb{Z}[\omega]$, so $\Lambda_1 = \mathcal{O}_K$ and $\Lambda_2 = \mathbb{Z}[\sqrt{-15}] = \mathcal{O}_2$, the conductor-2 order; $\mathcal{O}_2 = 2\mathcal{O}_K \sqcup C''$ with $C'' = \{\alpha \in \mathcal{O}_K : N(\alpha) \text{ odd}\}$, evaluated by Euler-factor removal at the two primes above 2 ($\chi_{-15}(2) = +1$); the exact track gives comb $= (6, 8, 3/2, 11/2)$ on the basis $\big(L_1, L_2, \zeta_K(2), L(\chi_5, 2)L(\chi_{-3}, 2)\big)$ and $\mathrm{EK}_4(\tau_0) = (5\sqrt{15}/\pi^3)\,\mathrm{comb}$, the right-hand side of Theorem 8.3; [V3] half-width 1.01×10^{-52}, mp agreement 1.24×10^{-60}. □

Theorems J and K. S3: both points lie on the certified vertical segment: $(3 + \mathrm{i}\sqrt{21})/6$ at $y = \sqrt{21}/6 \approx 0.764$ (also on leg B of `n4_p4_t4_cert.py`) and $(1 + \mathrm{i}\sqrt{21})/2$ at $y = \sqrt{21}/2$, the top endpoint of the segment, certified interior with margin 14.97. S4 is `cert0_n5_e78.py`: a $\mathbb{Q}(\sqrt{3})$ pair lock, minimal polynomial $X^2 + 1787904X + 84934656$ (discriminant $3 \cdot 1032192^2$), root separation 1.79×10^6 against lock widths 8.3×10^{-52} and 2.6×10^{-47}. S5 is `verify_P1_n5_e78.py` (73 checks): here the EK_4 lattices are *class* lattices, not orders — $T(\Lambda_1) = 1296\,T(\mathfrak{p}_6)$, $T(\Lambda_2) = 81\,T(\mathfrak{p}_3)$ for Theorem 8.4 and $T(\Lambda_1) = 16\,T(\mathfrak{p}_2)$, $T(\Lambda_2) = T(\mathcal{O}_K)$ for Theorem 8.5 — and with the four-class anchors of §8.2(iii) the combinations collapse to

$$\mathrm{comb}_J = -1296\,T(\mathfrak{p}_6) + 324\,T(\mathfrak{p}_3) = 36(A_3 - A_4) + 72\big(L_3 - L_4\big),$$
$$\mathrm{comb}_K = -16\,T(\mathfrak{p}_2) + 4\,T(\mathcal{O}_K) = 4(A_3 + A_4) + 8\big(L_3 + L_4\big),$$

with L_i the values $L\big(g(\varepsilon_2, \varepsilon_3), 3\big)$; EK_4 contributes the factors $5\sqrt{21}/(3\pi^3)$ and $5\sqrt{21}/\pi^3$ respectively, and the uniform Fricke relation of §8.2(iii) closes the assembly. The interval locks [V5]/[V6] have half-widths 5.09×10^{-52} and 1.70×10^{-52}; the mp residuals are 0 and 2.49×10^{-60}. □

8.2 Three methodological additions

(i) *One line, one certificate.* `n5_line_cert.py` certifies the entire vertical segment $\mathrm{Re}\tau = 1/2$, $0.702 \le \mathrm{Im}\tau \le \sqrt{21}/2$ in a single run: the anchor disk (one box, margin 0.29143, certified $|s_4| > 256$), the horizontal leg $y = 1$ (2 blocks, minimum margin 0.14759) and the vertical leg (6 blocks, minimum margin 7.29×10^{-3}; global minimum 0.0072856708 over the 8 blocks), with interval-certified interior-point verdicts at the three targets on the line (margins 3.16, 9.54, 14.97) and $23/23 + 5/5$ self-tests. The single certificate covers the CM points of Theorems 8.2, 8.3, 8.4 and 8.5 simultaneously — and also the point $\tau = (1 + \sqrt{-7})/2$ of Remark 1.1 ($s = -3969$), whose ordinate $\sqrt{7}/2 \approx 1.323$ lies inside the segment.

(ii) *The* $\mathbb{Q}(\sqrt{-15})$ *orbit trap.* For discriminant -15 the principal form is $x^2+xy+4y^2$, *not* x^2+15y^2: the latter, with $3x^2+5y^2$, belongs to the order of discriminant -60, and newforms assembled as $P \pm Q$ from that wrong orbit fail every Hecke and Fricke check. The correct construction uses the field form and the $\mathfrak{p}_2$-elements $\{c^2+15d^2 = 8n,\ c \equiv d\ (4)\}$: $a_1 = P+(\bar\omega/8)R$, $a_2 = P-(\bar\omega/8)R$; the genus character flips the non-principal part (also at the ramified primes $3, 5$, where the naive χ_5-twist formula fails); the $\sqrt{-15}$ parts cancel exactly and the coefficients are integral (Hecke/CM checks pass, Fricke ratio $+1$). Full details in the header of `verify_P1_n5_e6.py`.

(iii) *Class number four: the four-class anchor decomposition.* For $K = \mathbb{Q}(\sqrt{-21})$ the class group coincides with the genus group $(\mathbb{Z}/2)^2$, and the Dirichlet side of the lattice sums decomposes into the three genus characters with discriminant pairs $(-4, 21)$, $(-3, 28)$, $(-7, 12)$, with anchors

$A_4 = L(\chi_{-4},2)L(\chi_{21},2)$, $A_3 = L(\chi_{-3},2)L(\chi_{28},2)$, $A_7 = L(\chi_{-7},2)L(\chi_{12},2)$. The class-lattice sums are

$$\begin{aligned}
T(\mathcal{O}_K) &= \tfrac{1}{2}(z_K + A_4 + A_3 + A_7) + (L_1 + L_2 + L_3 + L_4),\\
T(\mathfrak{p}_2) &= \tfrac{1}{8}(z_K - A_4 - A_3 + A_7) + \tfrac{1}{4}(L_1 + L_2 - L_3 - L_4),\\
T(\mathfrak{p}_3) &= \tfrac{1}{18}(z_K - A_4 + A_3 - A_7) + \tfrac{1}{9}(L_1 - L_2 + L_3 - L_4),\\
T(\mathfrak{p}_6) &= \tfrac{1}{72}(z_K + A_4 - A_3 - A_7) + \tfrac{1}{36}(L_1 - L_2 - L_3 + L_4),
\end{aligned}$$

with $z_K = \zeta_K(2)$. The even-character values have the closed forms $L(\chi_{28},2) = 2\sqrt{7}\,\pi^2/49$ and $L(\chi_{21},2) = 8\sqrt{21}\,\pi^2/441$, certified by reduction to the finite integer identities $\sum_a \chi_{28}(a)a^2 = 448$ and $\sum_a \chi_{21}(a)a^2 = 168$; the Fricke relation $L'(g,0) = (42\sqrt{21}/\pi^3)\,L(g,3)$ (level 84, root number $+1$) is uniform over the four newforms; and in the final assembly z_K and A_7 both cancel (§8.1).

With Theorems 8.1–8.5, every entry of Samart's Table 6 that the machine can reach is proved: twenty-three of twenty-nine. The five remaining entries are blocked by the two obstructions analyzed in Section 10.

9 Conjectures beyond Table 6

9.1 Six new identities at Heegner points

The evaluation track of the machine (station S5) applies to any CM point, independently of the certification stations S3/S4. Applied to the Heegner points $\tau_D = (1+\sqrt{-D})/2$ with $D \in \{11, 19, 43, 67, 163\}$ — the class-number-one discriminants $D \equiv 3 \pmod 8$ with $D > 3$ (for $D = 3$ see Theorem 1.3), in which 2 is inert — and to $\tau_{27} = (1+3\sqrt{-3})/2$, attached to the order of conductor 3 in $\mathbb{Q}(\sqrt{-3})$, it produces a *uniform* answer, derived exactly from the lattice decomposition and independently recovered by integer-relation search. The answer involves a constant type that does not occur in Samart's library.

Definition 9.1 (The ray-class constants)**.** Let $K = \mathbb{Q}(\sqrt{-D})$ with D as above.

1. $M_D := L'(g_D, 0)$, where g_D is the newform of weight 3 and level D, CM by the principal Hecke character of K (class number one); $d_D := L'(\chi_{-D}, -1)$ as throughout.

2. $MU_D := \dfrac{(4D)^{3/2}}{4\pi^3}\big(L(\Psi_1,3) + L(\Psi_2,3)\big)$, where Ψ_1, Ψ_2 are the two conjugate Hecke characters of K of conductor (2), whose theta series are the level-$4D$ CM newform pair; by the functional equations $MU_D = \pm\big(M_{4D}^{(1)} + M_{4D}^{(2)}\big)$, the sign being determined by the root numbers of the pair.

3. (*New type*) For $D \in \{11, 19, 43, 67, 163\}$: $DU_D := \dfrac{3D^{3/2}}{2\pi^3}\, BU_D$, $BU_D := \dfrac{2B_1 - B_a - B_b}{2}$, where $B_r := \displaystyle\sum_{\alpha \in r + 2\mathcal{O}_K} N(\alpha)^{-2}$ are the partial zeta values of the ray classes modulo (2) and $1, a, b$ are representatives of the three classes; equivalently BU_D is the corresponding $L(2,\cdot)$-combination of the order-3 ray class characters of the conductor-2 order $\mathcal{O}_2$, whose class group is $\mathbb{Z}/3\mathbb{Z}$.

4. For $D = 27$: $M_{27} := L'(g_{27}, 0)$ for the level-27 newform CM by the order of discriminant -27; $DU_{27} := \dfrac{9\sqrt{3}}{2\pi^3}\, BU_{27}$, with BU_{27} the same combination of the partial zeta values of the ray

classes modulo (2) *in the conductor-3 order* $\mathcal{O}_3$ (the ring class group of the conductor-6 order is again $\mathbb{Z}/3\mathbb{Z}$); $DX_{27} := \dfrac{9\sqrt{3}}{2\pi^3}\,X_{27}$, $X_{27} := \displaystyle\sum_{\gamma\in\omega+\mathcal{O}_3} N(\gamma)^{-2}$, the partial zeta value of the coset of $\omega = (1+\sqrt{-3})/2$ in $\mathcal{O}_3$.

Conjecture 9.2 (The Heegner-point identities)**.** For $D \in \{11, 19, 43, 67, 163\}$,

$$n_4\big(s_4(\tau_D)\big) = \frac{10}{9D}\big(36M_D + 12MU_D + 3d_D + 8DU_D\big), \qquad \tau_D = \frac{1+\sqrt{-D}}{2}, \tag{11}$$

and

$$n_4\big(s_4(\tau_{27})\big) = \frac{10}{81}\big(12M_{27} + 4MU_{27} + 81d_3 - 27DX_{27} + 72DU_{27}\big), \qquad \tau_{27} = \frac{1+3\sqrt{-3}}{2}. \tag{12}$$

Remark 9.3 (The $D = 27$ identity simplifies)*.* The constant DX_{27} turns out *not* to be new: the evaluation machine gives

$$DX_{27} = \frac{56}{27}\,d_3$$

to 120 digits (the partial zeta X_{27} over the ramified prime 3 reduces to $L(\chi_{-3}, 2)$), so (12) is equivalently

$$n_4\big(s_4(\tau_{27})\big) = \frac{10}{81}\big(12M_{27} + 4MU_{27} + 25d_3 + 72DU_{27}\big).$$

The genuinely new constant in Conjecture 9.2 is DU_D.

The evidence is threefold. (i) *Exact derivation.* For $D \equiv 3$ (8) the EK_4 lattice pair is $\Lambda_1 = \mathcal{O}_K$, $\Lambda_2 = \mathcal{O}_2$ with $\mathcal{O}_2 = 2\mathcal{O}_K \sqcup (1 + 2\mathcal{O}_K)$ — two cosets, not four: $\omega + 2\mathcal{O}_K \not\subset \mathcal{O}_2$ — and $\mathrm{EK}_4(\tau_D) = (10y_D/\pi^3)\big({-T(\mathcal{O}_K)} + 4T(\mathcal{O}_2)\big)$, $y_D = \sqrt{D}/2$. The B/G-decomposition $T(\mathcal{O}_K) = 2\zeta_K(2) + 4L(g_D, 3)$ and $-T(\mathcal{O}_K) + 4T(\mathcal{O}_2) = \zeta_K(2) + \frac{8}{3}BU_D + 2L(g_D, 3) + \frac{16}{3}U_D$, with $U_D := L(\Psi_1, 3) + L(\Psi_2, 3)$, evaluates this exactly; conversion to the (M_D, MU_D, d_D, DU_D) units via the functional equations gives the displayed coefficients of (11) (for instance the M_D-coefficient: $\frac{5\sqrt{D}}{\pi^3}\cdot 2\cdot\frac{4\pi^3}{D^{3/2}} = \frac{40}{D} = \frac{10}{9D}\cdot 36$). For $D = 27$ the pair is $\mathcal{O}_3$, $\mathcal{O}_6 = 2\mathcal{O}_3 \sqcup (1+2\mathcal{O}_3)$ with $B(\mathcal{O}_3) = 6\zeta_K(2) - 2X_{27}$ (the unit group μ_6 and the ramified prime 3), giving $(40/27, 40/81, 10, -10/3, 80/9)$ on the basis $(M_{27}, MU_{27}, d_3, DX_{27}, DU_{27})$. (ii) *Integer-relation independence.* On the basis $[\mathrm{EK}_4, M_D, MU_D, d_D, DU_D]$ at 60 digits, PSLQ returns the single relation $[-9D, 360, 120, 30, 80]$ for each of the five discriminants — the same coefficients, found with no lattice input. (iii) *Control.* The same code with $D = 7$ (where 2 splits and no ray constants occur) reproduces Samart's Table 6 entry $\mathrm{EK}_4 = \frac{10}{7}(40M_7 + d_7)$ with residual 0 (`gen_conj_fit.py`); the self-checks of the shifted B/G-row machine (Euler-odd identities, $X(\omega) = X(2\omega)$ for $D = 27$) hold to 10^{-60} (`gen_conj_fit4.py`, `gen_conj_fit5.py`).

Remark 9.4 (Status)*.* Conjecture 9.2 is one certification short of a proof. All six parameters satisfy $|s_4(\tau_D)| > 256$ (Table 2), and a 40-digit scan of the segment $\mathrm{Im}\tau_D \le \mathrm{Im}\tau \le \mathrm{Im}\tau_D + 8$ of the vertical ray $\mathrm{Re}\tau = 1/2$ finds the minimum of $|s_4|$ at the bottom endpoint τ_D itself (`gen_conj_s4.py`); once the ray is certified by interval arithmetic — a one-leg instance of the `cert2` machine of §2.5.2 — the anchor Lemma 2.13 applies on a neighbourhood of τ_D and the exact evaluation above becomes a proof. We have not yet run that certification, and we do not know minimal polynomials of the parameters $s_4(\tau_D)$: PSLQ at 110 digits excludes degree ≤ 3 with coefficients $\le 10^8$ and degree 4 with coefficients $\le 10^6$, so station S4 is open. Curiously $s_4(\tau_D) \approx j(D) - 640$ with an error decaying like $\mathrm{e}^{-c\sqrt{D}}$ (2.2×10^{-4} at $D = 43$; 7.3×10^{-13} at $D = 163$, the $\mathrm{e}^{\pi\sqrt{163}}$ effect; `gen_conj_minpoly.py`).

D	$s_4(\tau_D)$	residual
11	$-33402.2734634215487007576553 8\ldots$	2.49×10^{-60}
19	$-885375.7826174678605529362693 4\ldots$	6.22×10^{-60}
27	$-12288639.9843344235156343818481 6\ldots$	2.49×10^{-60}
43	$-884736639.9997824076177230809381 9\ldots$	0
67	$-147197952639.9999986921557245470483 2\ldots$	4.98×10^{-60}
163	$-262537412640768639.99999999999926673\ldots$	9.96×10^{-60}

Table 2: The six Heegner-point parameters (60 digits; truncated) and the residuals $|n_4(s_4(\tau_D)) - \mathrm{RHS}|$; the PSLQ re-checks ($D \neq 27$) give residuals $\leq 1.4 \times 10^{-56}$. Values of the constants to 15 digits: $(M_{11}, MU_{11}, d_{11}, DU_{11}) = (0.257838994176722, 5.13771317570429, 2.64058735875153, 3.03409158565142)$; $(M_{163}, MU_{163}, d_{163}, DU_{163}) = (18.0996439429792, 272.773494341141, 109.694008111980, 203.758012415278)$; the full tables are in the repository.

Remark 9.5 (A constant type beyond Samart's library). Samart's constant library consists of the values M_N and d_k. The constant DU_D does not belong to it: PSLQ on $[DU_{11}, d_{11}, d_3, d_4, d_8]$ with coefficient bound 10^7 returns no relation. The arithmetic reason is that for $D \equiv 3$ (8) the prime 2 is inert and the conductor-2 order has class group $\mathbb{Z}/3\mathbb{Z}$, whose characters have order 3 — not quadratic Dirichlet characters. It is consistent with this picture that the Heegner-point entries of Samart's Table 6 all have $D \equiv 5, 7$ (8), with the single exception of the rational value $s = -144$ attached to $D = 3$: there the unit group μ_6 collapses the conductor-2 ring class number to 1, and no order-3 characters arise (Theorem 1.3 indeed uses only M_{12} and d_3). For $D \equiv 5$ (8) the conductor-2 ring class groups are 2-primary ($\mathbb{Z}/4\mathbb{Z}$ for $D = 5, 13$; $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ for $D = 21$), so the order-3 ray constants of Conjecture 9.2 are genuinely specific to $D \equiv 3$ (8), $D > 3$. Conjecture 9.2 is thus the first batch of identities *predicting* a new constant type rather than repackaging known ones.

9.2 An umbrella conjecture

We record the conjectural framework into which the results of this paper fit, and two refinements suggested by the present data. It is a standard expectation, implicit in [18] and made formal by Trieu's conditional theorem [30, Thm. 0.2]: for a three-variable polynomial P whose zero locus W_P has genus 1, with a wedge decomposition of the symbol xyz^3 [30, (0.8)], Beilinson's conjecture [21] implies $m(P) = m(\widetilde{P}) + a\,L'(E, -1)$ up to Bloch–Wigner dilogarithmic corrections, which in some cases simplify to Dirichlet L-values (the $K3$-surface sequel [31] is of the same shape). Thus every Mahler measure identity of Boyd–Samart type is expected to be an instance of Beilinson's conjecture for the corresponding motive, with the exact rational coefficients governed by Bloch–Kato [22]; see also [29] for the elliptic-polylogarithm picture and [19] for a systematic account. For the modular families of the present paper the special fibre is a singular K3 surface [25]: by Shioda–Inose [27] its L-function splits off a weight-3 CM newform (the M-terms of our identities), while the Dirichlet terms fall on the proved line of the K_3-theory of imaginary quadratic fields [28]. Fei's theorem [4, Thm. 0.2] — 179 identities of the same two-term shape, covering 23 of his 25 Landau–Ginzburg families, in which the newform levels satisfy $-D/d$ a square and the Dirichlet discriminants d' satisfy $d' \mid D$ — is the corresponding theorem-level catalogue; and the 25 families known today do not exhaust the rational weight-3 CM newforms classified by Schütt [20]. But the literature contains no family-level conjecture predicting the discriminant structure (which conductors occur) or the coefficient structure (how large the denominators are). Theorems A–K are unconditional CM instances of this framework, and Conjecture 9.2 is its first evidence beyond the 2-primary world; we propose the following two conjectures to organize them. They are not restatements of

Beilinson's conjecture: their predictions (discriminant structure, denominators, no-formula cases) are not implied by it.

The mechanism expected to produce the shape below is that the holomorphic Mahler function $\widetilde{\mathrm{m}}(c(\tau))$ is a Beilinson regulator pairing of a motivic class attached to the family with the Deninger cycle — cf. [30], where this is proved for single polynomials. We do not attempt to make the motivic clause precise here; the conjecture records only its falsifiable arithmetic consequence.

Conjecture 9.6 (CM specialization: shape and discriminant structure)**.** Let P_c be a Laurent-polynomial family with a Hauptmodul parametrization $c = c(\tau)$ by a genus-0 congruence group Γ, whose fibres are elliptic curves or K3 surfaces. Then for every CM point τ_0 of discriminant D with $c(\tau_0) \in \bar{\mathbb{Q}}$, the value $\mathrm{m}(P_{c(\tau_0)})$ lies in the $\mathbb{Q}$-span of the values $L'(g, 0)$ and $L(\chi, 2)$, where g runs over the weight-3 newforms CM by Hecke characters of the orders of discriminant D, and χ over the ray class characters of those orders; the levels and conductors are explicitly determined by the level of Γ and by D.

The new content: (1) the occurring characters are delimited a priori — Fei's constraint ($d \mid D$, D/d a square) is the genus-theoretic, 2-primary part, while Conjecture 9.2 shows the odd part of the ring class group entering through order-3 ray characters, and the conjecture predicts the same for every order; (2) completeness: every algebraic singular modulus in the c-image carries a formula; (3) a Galois-orbit form: via the determinant machine of [7], the single-point identities are the $n = 1$ slices of $n \times n$ determinant identities over CM Galois orbits.

Conjecture 9.7 (Denominators are controlled by local data)**.** In every identity of Conjecture 9.6, the reduced denominators of the rational coefficients divide an explicit integer formed from the Tamagawa numbers (component groups) of the special fibre at its bad primes, the unit index of the order, and the 2-torsion of its class group.

This is the family analogue of the Bloch–Grayson modification of Beilinson's conjecture [23] (regulators taken on integral elements; numerically substantiated for hyperelliptic curves in [26]) — the Mahler slice of Bloch–Kato; Boyd already expected the Bloch–Kato theory to predict the exact rational factor [24]. Its predictions: (i) a priori denominator bounds for the unproved entries, falsifiable at scale by integer-relation search; (ii) a no-formula criterion — if the special fibre's transcendental motive is not of CM type, no identity of this shape exists — consonant with [20] and with Fei's remark; (iii) the same local integer controls the denominators of the higher-derivative versions $L^{(k)}$.

Remark 9.8 (First data). Theorems A–K supply twelve proved identities, with overall denominators 3, 4, 7, 8, 14, 15, 16, 21, 24, 28, 48. Conjecture 9.2 supplies six conjectured ones: $9D$ for the five discriminants of (11), and 81. We caution that the factor D in $9D$ is a normalization artifact: it enters through the $D^{3/2}$ factor converting $L(g_D, 3)$ into M_D (in the natural $(\zeta_K(2), L(g_D, 3), BU_D)$ units no D-denominator appears), so it carries no Tamagawa information. The nontrivial content for Conjecture 9.7 is the residual factor 9 (and 81).

10 The boundary of the method

We collect the precise limits of the machine, all visible already in Table 1.

10.1 Two-dimensional critical images

The decisive invariant is the dimension of the critical image (Remark 2.4). In the n_2 family the image is the slit $[0, 64]$, which does not separate $\mathbb{C}$, and every conjectured entry was reachable: the family is complete [5, 10, 11]. In the n_4 family the image is the astroid disc (4), which has interior. A CM parameter with $c(\tau_0)$ in the interior of Ast lies inside an open bad region of the τ-plane (open mapping), no certified path can approach it, and Theorem 2.10 is silent. The extreme case is the refuted conjecture $n_4(81) = 40M_7$ [6]: $c = 3$ is an interior lattice point of the astroid, the conjectured value equals the exact EK evaluation at the attached CM point on a *wrong sheet*, and direct torus integration gives the different value $n_4(81) = 4.1655349907533676508(5)$. Entries #4 and #5 of Table 1 have $c(\tau_0)$ inside Ast ($s \approx -2.45$ and $s \approx -6.17$ respectively) and are blocked by the same mechanism; entry #5 is moreover exterior (see the next subsection).

10.2 The wrong sheet below $\mathrm{Im}\tau = 1/\sqrt{2}$

Even where the critical image is avoided, the EK series itself can fail. A systematic scan (`diag_n4_astroid.py`) shows $\mathrm{EK}_4(\tau) \neq 4\mathrm{m}(P + c(\tau))$ *at every tested point* below the line $\mathrm{Im}\tau = 1/\sqrt{2}$ — including points with $c(\tau)$ real and larger than 4, where the ${}_5F_4$ evaluation converges and the Mahler side is perfectly smooth (e.g. a difference of 11.4 at $\tau = 0.3\mathrm{i}$). The difference grows continuously from 0 at the line; there is no jump. The mechanism is the same wrong-sheet phenomenon analyzed for the n_2 family in [5, Remark 5.6]: the single-valued series EK_4 follows a companion branch of $\widetilde{\mathrm{m}}$ once the branch locus is crossed. The propagation of Theorem 2.10 cannot cross, because the identity theorem fixes $\Psi \equiv 0$ only on the anchor's component W: along the imaginary axis the point $\tau = \mathrm{i}/\sqrt{2}$ (where $c = 4$, a cusp of Ast) separates W from the lower component W', and on W' the difference $H = F - G$ is non-constant. Entries #3, #9, and #10 of Table 1 (pure imaginary, $y = 0.612$, 0.154, and 0.463, with $c > 4$ real in each case) are blocked at this *series* level: the conjectured identity — a statement about the true Mahler measure, which Samart's PSLQ evidence supports — has no valid series input for station S5, and proving it requires a corrected expression, e.g. a modified Mahler measure of Samart–Tao $\widetilde{n}$-type [7], whose design principle (choose the linear combination so that the non-closed Deninger path [18] becomes a meaningful pairing) is compatible with stations S2/S4 of the machine.

10.3 Open directions

- *The formerly ready entries* #1, #2, #6, #7, #8 are now proved as Theorems 8.1–8.5 (Section 8); Table 6 has no reachable entries left.
- *The blocked entries* #3, #4, #5, #9, #10: a corrected (wrong-sheet-free) formula is needed before the machine has anything to certify; the $\widetilde{n}$ framework of [7] is the natural candidate.
- *Samart's three-modular-L-value hypothesis* $s = 16 + 1600\sqrt[3]{2} - 1280\sqrt[3]{4}$ [2]: the conjectured value involves three modular L-values and no attached CM point in the s_4-image is known; the machine's station S1 has no input.
- *The Boyd-lineage entries*: the two-variable conjectures of Boyd's tables (e.g. conductor-14 and related families) have no known CM evaluation, so the machine does not apply; regulator-based approaches [18, 19], and in particular deconditioning of the currently conditional identities in the Brunault–Zudilin framework [19], remain the promising route.

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